Mathematical Statistics I

Lecture notes based on Kim, *Mathematical Statistics*, Ch. 1–5

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Chapter 1. Probability Distributions

1.1 Random experiments and probability

Motivation. Statistics infers properties of a population from random samples, so it needs a precise model of randomness. Kolmogorov's three axioms are that model; every rule of probability in these notes is derived from them.

A **random experiment** has outcomes that cannot be predicted in advance; its **sample space** S is the set of all outcomes, and **events** are subsets of S. When all N outcomes of a finite S are equally likely, $P(A) = \#A/N$ (classical probability); in general a probability is any assignment obeying Kolmogorov's axioms:

(A1) $P(A) \geq 0$ for every event A;

(A2) $P(S) = 1$;

(A3) (countable additivity) if $A_1, A_2, \ldots$ are pairwise disjoint, $P(\bigcup_i A_i) = \sum_i P(A_i)$.

Theorem 1.1 (Basic properties)

(a) $P(\emptyset) = 0$, and finite additivity holds.

(b) $P(A^c) = 1 - P(A)$, and $0 \leq P(A) \leq 1$.

(c) (Monotonicity) $A \subseteq B \Rightarrow P(A) \leq P(B)$, and $P(B \setminus A) = P(B) - P(A)$.

(d) (Addition rule) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; more generally, inclusion–exclusion

$$P\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n-1} P(A_1 \cap \cdots \cap A_n).$$

(e) (Countable subadditivity) $P(\bigcup_i A_i) \leq \sum_i P(A_i)$.

Proof. (a) Apply (A3) to $S \cup \emptyset \cup \emptyset \cup \cdots$: $1 = 1 + \sum P(\emptyset)$ forces $P(\emptyset) = 0$; padding a finite union with empty sets gives finite additivity.

(b) $S = A \cup A^c$ disjointly.

(c) $B = A \cup (B \setminus A)$ disjointly.

(d) $A \cup B = A \cup (B \setminus (A \cap B))$ disjointly and $P(B \setminus (A \cap B)) = P(B) - P(A \cap B)$ by (c); the general formula follows by induction on n, applying the two-set rule to $(A_1 \cup \cdots \cup A_{n-1}) \cup A_n$.

(e) Put $B_1 = A_1$, $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$; these are disjoint with the same union and $B_n \subseteq A_n$. ■

Theorem 1.2 (Continuity of probability)

If $A_1 \subseteq A_2 \subseteq \cdots$ then $P(\bigcup_n A_n) = \lim_n P(A_n)$; if $B_1 \supseteq B_2 \supseteq \cdots$ then $P(\bigcap_n B_n) = \lim_n P(B_n)$.

Proof. With $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1}$ (disjoint, same union),

$$P\Big(\bigcup_n A_n\Big) = \sum_k P(C_k) = \lim_n \sum_{k \leq n} P(C_k) = \lim_n P(A_n).$$

For decreasing B_n, apply this to $A_n = B_n^c$ and take complements. ■

Example 1.1

Let $P((a,b]) = b-a$ for subintervals of $[0,1]$ (the uniform probability). For a point b, $\{b\} = \bigcap_n (b - \frac{1}{n}, b]$, so $P(\{b\}) = \lim_n \frac{1}{n} = 0$: every single point has probability zero, although some point always occurs. Likewise $P([a,b]) = P((a,b)) = b-a$.

Example 1.2

Roll three fair dice. $P(\text{at least one 6}) = 1 - P(\text{no 6}) = 1-(5/6)^3 = 91/216 \approx 0.42$. Inclusion–exclusion gives the same: $3 \cdot \frac{1}{6} - 3 \cdot \frac{1}{36} + \frac{1}{216} = \frac{91}{216}$.

1.2 Conditional probability and independence

Motivation. Data change our probabilities: conditional probability restricts attention to outcomes consistent with what has been observed. Bayes's rule reverses the conditioning, and independence is the special case where observing one event tells nothing about another.

Definition (Conditional probability)

For $P(A) > 0$, $P(B \mid A) = P(A \cap B)/P(A)$. As a function of B it satisfies (A1)–(A3), so it is a probability on the reduced sample space A.

Theorem 1.3

(a) (Multiplication) $P(A \cap B) = P(B \mid A)P(A)$, and $P(A_1 \cap \cdots \cap A_n) = P(A_1)P(A_2 \mid A_1) \cdots P(A_n \mid A_1 \cap \cdots \cap A_{n-1})$.

(b) (Total probability) If $A_1, A_2, \ldots$ partition S with $P(A_i) > 0$, then $P(B) = \sum_i P(B \mid A_i)P(A_i)$.

(c) (Bayes) In the same setting, with $P(B) > 0$,

$$P(A_j \mid B) = \frac{P(B \mid A_j)P(A_j)}{\sum_i P(B \mid A_i)P(A_i)}.$$

Proof. (a) is the definition, iterated.

(b) $B = \bigcup_i (B \cap A_i)$ disjointly; apply (A3) and (a).

(c) $P(A_j \mid B) = P(B \cap A_j)/P(B)$; expand the numerator by (a) and the denominator by (b). ∎

$P(A_j)$ are **prior** and $P(A_j \mid B)$ **posterior** probabilities.

Example 1.3 (Diagnostic test)

A disease has prevalence 1%; a test detects it with probability 0.95 and gives false positives with probability 0.05. Then

$$P(\text{disease} \mid +) = \frac{0.95 \cdot 0.01}{0.95 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0095}{0.059} \approx 0.16.$$

A positive result raises the probability only from 1% to 16% because healthy people vastly outnumber sick ones.

Example 1.4 (Three machines)

Machines A, B, C make $50\%, 30\%, 20\%$ of output with defect rates $2\%, 3\%, 5\%$. Then $P(\text{defective}) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.029$ and $P(\text{C} \mid \text{defective}) = 0.010/0.029 \approx 0.34$.

Definition (Independence)

A, B are independent if $P(A \cap B) = P(A)P(B)$ (equivalently $P(B \mid A) = P(B)$ when $P(A) > 0$). $A_1, \ldots, A_n$ are (mutually) independent if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every subcollection.

If A, B are independent, so are A, B^c (and A^c, B^c): $P(A \cap B^c) = P(A) - P(A)P(B) = P(A)P(B^c)$. Pairwise independence does not imply mutual independence:

Example 1.5

Toss two fair coins: $A = \{\text{first H}\}$, $B = \{\text{second H}\}$, $C = \{\text{the two agree}\}$. Each has probability $\frac{1}{2}$ and each pair intersects with probability $\frac{1}{4}$, but $P(A \cap B \cap C) = \frac{1}{4} \neq \frac{1}{8}$.

Example 1.6 (Parallel system)

A device works if at least one of three independent components works; each fails with probability 0.1. $P(\text{works}) = 1 - 0.1^3 = 0.999$.

1.3 Random variables and their distributions

A **random variable** X is a real-valued function on S; events such as $\{X \le x\} = \{s : X(s) \le x\}$ inherit probabilities, and the probabilities $P(X \in A)$ form the **distribution** of X.

- **Discrete** X takes values $x_1, x_2, \ldots$; its probability mass function (pmf) $f(x) = P(X = x)$ satisfies $f \ge 0$, $f = 0$ off $\{x_k\}$, $\sum_x f(x) = 1$, $P(a \le X \le b) = \sum_{a \le x \le b} f(x)$.
- **Continuous** X has a probability density function (pdf) $f \ge 0$ with $\int f = 1$ and $P(a \le X \le b) = \int_a^b f(x)\,dx$. Then $P(X = a) = 0$ and $P(a \le X \le a + \Delta) \approx f(a)\Delta$, so $f(a)$ is a probability per unit length, not a probability.

We write $X \sim f$; the indicator $I_A(x)$ gives compact formulas such as $f(x) = 2x I_{[0,1]}(x)$.

Example 1.7

Toss a fair coin until the first head; the number of tosses X has $f(k) = (1/2)^k$, $k = 1, 2, \ldots$, and $\sum_k f(k) = 1$. Two tosses give $Y =$ number of heads with $f_Y(0) = \frac{1}{4}$, $f_Y(1) = \frac{1}{2}$, $f_Y(2) = \frac{1}{4}$.

Example 1.8

If $f(x) = cx(2 - x)$ on $[0, 2]$, then $\int_0^2 x(2 - x)\,dx = \frac{4}{3}$, so $c = \frac{3}{4}$ and $P(X \le \frac{1}{2}) = \frac{3}{4}\int_0^{1/2}(2x - x^2)\,dx = \frac{3}{4}(\frac{1}{4} - \frac{1}{24}) = \frac{5}{32}$.

1.4 Expectation, variance and moments

Motivation. A distribution is a whole function, but we usually summarize it by a few numbers: its center (mean) and spread (variance). Higher moments describe shape, and expectations of functions of X are the building blocks of estimators.

Definition (Expectation)

$E[g(X)] = \sum_x g(x)f(x)$ or $\int g(x)f(x)\,dx$, defined when the sum or integral converges absolutely. $\mu = E(X)$ is the mean (the distribution's center of mass), $\operatorname{Var}(X) = E[(X-\mu)^2]$ the variance and $\operatorname{sd}(X) = \sqrt{\operatorname{Var} X}$ the standard deviation.

Theorem 1.4 (Properties of expectation)

(a) $E(aX+b) = aE(X)+b$.
(b) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E g_1(X) + c_2 E g_2(X)$.
(c) $g_1 \le g_2 \Rightarrow E g_1(X) \le E g_2(X)$.

Proof. Linearity and monotonicity of sums and integrals. ■

Theorem 1.5 (Variance)

(a) $\operatorname{Var}(aX+b) = a^2 \operatorname{Var} X$.
(b) $\operatorname{Var} X = E(X^2) - (EX)^2$.

Proof. (a) $aX + b - E(aX+b) = a(X-\mu)$, so $\operatorname{Var}(aX+b) = E[a^2(X-\mu)^2]$.
(b) $E(X-\mu)^2 = E(X^2) - 2\mu EX + \mu^2 = E(X^2) - \mu^2$. ■

The **standardized** variable $Z = (X-\mu)/\sigma$ has $EZ = 0$, $\operatorname{Var} Z = 1$. A mean need not exist: for $f(k) = \frac{1}{k(k+1)}$, $k \ge 1$, $\sum k f(k) = \sum \frac{1}{k+1} = \infty$.

Example 1.9

For the uniform density on $[0,1]$, $\mu = \frac{1}{2}$ and $\sigma^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$. For the triangular density $f(x) = 4x$ on $[0, \frac{1}{2}]$ and $4 - 4x$ on $[\frac{1}{2}, 1]$ (same mean), $EX^2 = \frac{7}{24}$ and $\sigma^2 = \frac{7}{24} - \frac{1}{4} = \frac{1}{24}$: the triangle is half as spread.

Example 1.10

With two tosses and a prize of X^2 (in thousands of won) for X heads, $E(X^2) = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = 1.5$, so the expected prize is 1,500 won.

1.5 Distribution functions and generating functions

The **cumulative distribution function** (cdf) is $F(x) = P(X \le x)$. For discrete X it is a step function with jumps $f(x_k)$ at the x_k; for continuous X, $F(x) = \int_{-\infty}^{x} f$ and $F'(x) = f(x)$ where f is continuous.

Theorem 1.6 (Properties of a cdf)

(a) F is nondecreasing.
(b) $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to\infty} F(x) = 1$.
(c) F is right-continuous.
(d) $F(a-) = P(X < a)$, hence $F(a) - F(a-) = P(X = a)$.

Proof. (a) $\{X \le x_1\} \subseteq \{X \le x_2\}$.

(b) $\{X \le -k\} \downarrow \emptyset$ and $\{X \le k\} \uparrow S$; apply Theorem 1.2.

(c) $\{X \le x + \frac{1}{n}\} \downarrow \{X \le x\}$.

(d) $\{X \le a - \frac{1}{n}\} \uparrow \{X < a\}$. ■

Conversely every function with (a)–(c) is the cdf of some random variable.

Example 1.11 (A mixed distribution)

A commuter flips a coin: on heads they wait for a shuttle that comes every 10 minutes, so the trip takes $10 + U$ minutes with $U \sim U(0, 10)$; on tails they walk, which takes exactly 15 minutes. The trip time X has

$$F(x) = \tfrac{1}{2}P(10 + U \le x) + \tfrac{1}{2}I(x \ge 15) = \begin{cases} 0, & x < 10, \\ (x-10)/20, & 10 \le x < 15, \\ x/20, & 15 \le x < 20, \\ 1, & x \ge 20, \end{cases}$$

with a jump of $\frac{1}{2}$ at 15: neither discrete nor continuous.

Probability generating function. For X with values in $\{0, 1, 2, \dots\}$, $G(s) = E(s^X) = \sum_x s^x f(x)$ for $|s| < 1$. Since a power series determines its coefficients, $f(k) = G^{(k)}(0)/k!$ and G determines the distribution.

Moment generating function. If $E(e^{tX}) < \infty$ for all t in some $(-h, h)$, $h > 0$, the mgf is $M(t) = E(e^{tX})$; $m_k = E(X^k)$ is the kth moment.

Theorem 1.7 (Properties of the mgf)

If M exists on $(-h, h)$:
(a) all moments exist, $E(X^k) = M^{(k)}(0)$ and $M(t) = \sum_{k\ge0} \frac{E(X^k)}{k!} t^k$ near 0;
(b) if $M_X = M_Y$ on a neighborhood of 0, then X and Y have the same distribution.

Proof of (a). Fix $|t| < h$. Since $|tx|^k/k! \le e^{|tx|} \le e^{tx} + e^{-tx}$, $|t|^k E|X|^k/k! \le M(t) + M(-t) < \infty$, so all moments exist. By Taylor's theorem with integral remainder,

$$\Big|e^{tx} - \sum_{k=0}^{n-1} \frac{(tx)^k}{k!}\Big| = \Big|\frac{(tx)^n}{(n-1)!}\int_0^1 (1-u)^{n-1} e^{utx}\,du\Big| \le \frac{|tx|^n}{n!} e^{|tx|}.$$

Since $\frac{|tx|^n}{n!} = 2^{-n}\frac{|2tx|^n}{n!} \le 2^{-n}e^{2|tx|}$, the right side is at most $2^{-n}e^{3|tx|} \le 2^{-n}(e^{3tx} + e^{-3tx})$. Taking expectations, for $|3t| < h$,

$$\Big|M(t) - \sum_{k<n} \frac{E(X^k)}{k!} t^k\Big| \le 2^{-n}\{M(3t) + M(-3t)\} \to 0.$$

So M equals its power series near 0; differentiating term by term gives $M^{(k)}(0) = E(X^k)$.

(b) is proved with characteristic functions and is beyond these notes. ■

Existence of all moments does not imply existence of the mgf (the lognormal distribution is a counterexample).

Example 1.12

For the coin-until-head variable of Example 1.7: $G(s) = \sum_k (s/2)^k = \frac{s}{2-s}$, and $M(t) = G(e^t) = \frac{e^t/2}{1-e^t/2}$ for $t < \log 2$. Then $M'(0) = 2$: on average two tosses.

Example 1.13 (Standard exponential)

For $f(x) = e^{-x}$, $x > 0$: $M(t) = \int_0^\infty e^{-(1-t)x}\,dx = (1-t)^{-1} = \sum_k t^k$ for $|t| < 1$, so $E(X^k) = k!$. (Also directly by repeated integration by parts.)

Example 1.14 (Recovering a distribution from moments)

Suppose the mgf exists and $E(X^k) = 3^k k!$. Then $M(t) = \sum_k (3t)^k = (1-3t)^{-1}$ for $|t| < \frac{1}{3}$. Since $\frac{1}{3}e^{-x/3}$ $(x > 0)$ has exactly this mgf, uniqueness shows X has this density.

Cumulants. $C(t) = \log M(t) = \sum_{r\geq 1} c_r \frac{t^r}{r!}$ defines the cumulants c_r. Expanding $\log(1+A) = A - \frac{A^2}{2} + \frac{A^3}{3} - \cdots$ with $A = m_1 t + m_2 \frac{t^2}{2!} + \cdots$ and collecting powers of t gives $c_1 = m_1 = \mu$ and $c_2 = m_2 - m_1^2 = \sigma^2$. For the standard exponential, $C(t) = -\log(1-t) = \sum_r t^r/r$, so $c_r = (r-1)!$; for Example 1.14, $c_r = (r-1)!3^r$.

1.6 Inequalities

Motivation. In statistics we rarely know a distribution exactly, yet we need to bound probabilities and expectations. Markov, Chebyshev and Jensen give such bounds from moments alone and are used in every consistency proof.

Theorem 1.8 (Jensen)

If ϕ is convex on an interval I containing the values of X, and EX exists, then $\phi(EX) \leq E\phi(X)$.

Proof for twice differentiable ϕ. Let $\mu = EX \in I$. Taylor's theorem gives $\phi(x) = \phi(\mu) + \phi'(\mu)(x-\mu) + \frac{1}{2}\phi''(\xi)(x-\mu)^2$ with ξ between x and μ; since $\phi'' \geq 0$, $\phi(x) \geq \phi(\mu) + \phi'(\mu)(x-\mu)$ for all $x \in I$. Take expectations. (For general convex ϕ use a supporting line at μ.) ■

E.g. $(EX)^2 \leq E(X^2)$ and $E\log X \leq \log EX$ for $X > 0$.

Theorem 1.9 (Lyapunov)

If $E|X|^s < \infty$ and $0 < r < s$, then $(E|X|^r)^{1/r} \leq (E|X|^s)^{1/s}$.

Proof. For $p = s/r > 1$, $\phi(y) = y^p I(y \geq 0)$ is convex. $Y = |X|^r$ is integrable, since $|X|^r \leq I(|X| \leq 1) + |X|^s I(|X| > 1) \leq 1 + |X|^s$. Jensen gives $(EY)^p \leq E(Y^p) = E|X|^s$. ■

In particular, if the sth moment exists, all lower moments exist.

Theorem 1.10 (Markov and Chebyshev)

(a) If $E|Z|^r < \infty$ $(r > 0)$, then $P(|Z| \geq k) \leq E|Z|^r/k^r$ for every $k > 0$.
(b) If $\operatorname{Var} X < \infty$, then $P(|X - EX| \geq k) \leq \operatorname{Var} X/k^2$.

Proof. (a) $P(|Z| \geq k) = E\, I(|Z| \geq k)$, and pointwise $I(|Z| \geq k) \leq (|Z|/k)^r I(|Z|/k \geq 1) \leq |Z|^r/k^r$.
(b) Take $Z = X - EX$ and $r = 2$. ■

Example 1.15

Whatever the distribution, $P(|X - \mu| \geq 6\sigma) \leq 1/36 \approx 0.028$; for a normal population the exact value is about 2×10^{-9}. Chebyshev is universal but crude.

Theorem 1.11 (Zero variance)

If $\operatorname{Var} X = 0$ then $P(X = \mu) = 1$.

Proof. $\{|X - \mu| > 0\} = \bigcup_n \{|X - \mu| \geq \frac{1}{n}\}$, and by Chebyshev each event has probability at most $\operatorname{Var} X \cdot n^2 = 0$. Countable subadditivity gives $P(|X - \mu| > 0) = 0$. ■

1.7 Selected exercises

Motivation. These are the course's exercises for Chapter 1 (numbered as there). They concern the fine structure of distribution functions, how moments control tails, a basic moment inequality, and the sharp one-sided form of Chebyshev's inequality.

Exercise 1.7 (Jumps of a distribution function)

Show that the set of discontinuity points of a cdf F is countable.

Solution.

Step 1. By monotonicity the left limit $F(x-)$ exists at every x, and $F(x-) \leq F(x)$. Since F is right-continuous (Theorem 1.6), F is discontinuous at x iff $F(x) - F(x-) > 0$. Hence the set D of discontinuities is $\bigcup_{k \geq 1} D_k$ with $D_k = \{x : F(x) - F(x-) > 1/k\}$.

Step 2. Let $x_1 < \cdots < x_m$ be points of D_k. For $i \geq 2$, $F(x_i-) \geq F(x_{i-1})$ by monotonicity, so the jumps do not overlap and

$$\sum_{i=1}^{m} \{F(x_i) - F(x_i-)\} \leq F(x_m) - F(x_1-) \leq 1.$$

Each term exceeds $1/k$, so $m < k$. Thus D_k has fewer than k points.

Step 3. D is a countable union of finite sets, hence countable. (Equivalently, X has at most countably many atoms, since $P(X = x) = F(x) - F(x-)$.) ■

Exercise 1.10 (Moments and tails)

Let X be a random variable and $k > 0$.
(a) Show that $E|X|^k < \infty$ implies $n^k P(|X| > n) \to 0$ as $n \to \infty$.
(b) Show by an example that the converse of (a) fails.
(c) Show that if $n^\alpha P(|X| > n) \to 0$ for some $\alpha > 0$, then $E|X|^\beta < \infty$ for every $0 < \beta < \alpha$.

Solution.

(a) On the event $\{|X| > n\}$ we have $n^k < |X|^k$, so $n^k P(|X| > n) \le E[|X|^k I(|X| > n)]$. The right side is the tail beyond n of the convergent sum or integral defining $E|X|^k$ (for instance $\int_{|x|>n} |x|^k f(x)\,dx$), which tends to 0 as $n \to \infty$.

(b)

Step 1. Let X have density $f(x) = c/(x^2 \log x)$ for $x > e$, where $c^{-1} = \int_e^\infty \frac{dx}{x^2 \log x} \le \int_e^\infty \frac{dx}{x^2} < \infty$. Then $EX = c \int_e^\infty \frac{dx}{x \log x} = c\,[\log\log x]_e^\infty = \infty$.

Step 2. For $n > e$, since $\log x \ge \log n$ on (n, ∞),

$$nP(X > n) = n \int_n^\infty \frac{c\,dx}{x^2 \log x} \le \frac{cn}{\log n} \int_n^\infty \frac{dx}{x^2} = \frac{c}{\log n} \to 0.$$

So the converse fails for $k = 1$; for general k use $Y = X^{1/k}$, for which $EY^k = EX = \infty$ and $n^k P(Y > n) = n^k P(X > n^k) \le c/(k \log n) \to 0$.

(c)

Step 1. A convergent sequence is bounded, so there is C with $P(|X| > n) \le C n^{-\alpha}$ for all integers $n \ge 1$.

Step 2. Split Ω into $\{|X| \le 1\}$ and the dyadic shells $\{2^j < |X| \le 2^{j+1}\}$, $j \ge 0$. On the shell, $|X|^\beta \le 2^{(j+1)\beta}$, so

$$E|X|^\beta \le 1 + \sum_{j=0}^{\infty} 2^{(j+1)\beta} P(|X| > 2^j) \le 1 + C 2^\beta \sum_{j=0}^{\infty} 2^{-j(\alpha-\beta)} < \infty,$$

since $\alpha - \beta > 0$ makes the geometric series converge. ■

Exercise 1.13 (The c_r-inequality)

Show that for $r > 0$,

$$E|X+Y|^r \le c_r E|X|^r + c_r E|Y|^r, \qquad c_r = \begin{cases} 1, & 0 < r \le 1, \\ 2^{r-1}, & r > 1. \end{cases}$$

Solution. It suffices to prove $|a+b|^r \le c_r(|a|^r + |b|^r)$ for real a, b and then take expectations (Theorem 1.4(c)). Since $|a+b| \le |a| + |b|$ and $t \mapsto t^r$ is increasing on $[0, \infty)$, it is enough to bound $(|a| + |b|)^r$; assume $s = |a| + |b| > 0$.

Step 1. ($0 < r \le 1$.) Put $t = |a|/s \in [0, 1]$. Since $u^r \ge u$ for $u \in [0, 1]$, $t^r + (1-t)^r \ge t + (1-t) = 1$. Multiplying by s^r gives $(|a| + |b|)^r \le |a|^r + |b|^r$.

Step 2. ($r > 1$.) The function $u \mapsto u^r$ is convex on $[0, \infty)$, so

$$\Big(\frac{|a| + |b|}{2}\Big)^r \le \frac{|a|^r + |b|^r}{2}, \qquad \text{i.e.} \qquad (|a| + |b|)^r \le 2^{r-1}(|a|^r + |b|^r).$$

In particular $E|X|^r, E|Y|^r < \infty$ imply $E|X+Y|^r < \infty$. ■

Exercise 1.17 (One-sided Chebyshev inequality)

Let $EX = 0$ and $\operatorname{Var} X = \sigma^2$. Show that

$$P(X \geq x) \leq \frac{\sigma^2}{\sigma^2 + x^2} \quad (x > 0), \qquad P(X \geq x) \geq \frac{x^2}{\sigma^2 + x^2} \quad (x < 0).$$

Solution.

(a) Let $x > 0$ and $b \geq 0$. On $\{X \geq x\}$ we have $X + b \geq x + b > 0$, hence $(X + b)^2 \geq (x + b)^2$. By Theorem 1.10(a) with $r = 2$,

$$P(X \geq x) \leq P(|X + b| \geq x + b) \leq \frac{E(X + b)^2}{(x + b)^2} = \frac{\sigma^2 + b^2}{(x + b)^2}.$$

The choice $b = \sigma^2/x$ minimizes the right side and gives

$$\frac{\sigma^2 + \sigma^4/x^2}{(x + \sigma^2/x)^2} = \frac{\sigma^2(x^2 + \sigma^2)/x^2}{(x^2 + \sigma^2)^2/x^2} = \frac{\sigma^2}{\sigma^2 + x^2}.$$

(b) Let $x < 0$. Then $-X$ has mean 0 and variance σ^2, and $-x > 0$, so by (a)

$$P(X \geq x) = 1 - P(-X > -x) \geq 1 - P(-X \geq -x) \geq 1 - \frac{\sigma^2}{\sigma^2 + x^2} = \frac{x^2}{\sigma^2 + x^2}.$$

Both bounds are sharp: for $x > 0$, the two-point law with $P(X = x) = \frac{\sigma^2}{\sigma^2 + x^2}$ and $P(X = -\sigma^2/x) = \frac{x^2}{\sigma^2 + x^2}$ has mean 0, variance σ^2 and attains equality in (a). ■

Chapter 2. Multivariate Random Variables

2.1 Joint and marginal distributions

A pair (X, Y) is described by a **joint pmf** $f(x, y) = P(X = x, Y = y)$ in the discrete case, or a **joint pdf** $f \geq 0$ with $P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$ in the continuous case. In both cases $f \geq 0$ and f sums or integrates to 1.

Definition (Marginal distributions)

$f_1(x) = \sum_y f(x, y)$ or $\int f(x, y)\, dy$, and $f_2(y) = \sum_x f(x, y)$ or $\int f(x, y)\, dx$.

Marginals are obtained by "summing out" the other variable; in a table they are the row and column totals. The joint distribution determines the marginals, but not conversely.

The **joint cdf** is $F(x, y) = P(X \leq x, Y \leq y)$. For continuous (X, Y), $f = \partial^2 F / \partial x \partial y$ at continuity points. The marginal cdfs are limits: $F_1(x) = \lim_{y \to \infty} F(x, y)$.

Theorem 2.1

(a) F is nondecreasing in each argument, and for $x_1 < x_2$, $y_1 < y_2$,

$$P(x_1 < X \leq x_2,\ y_1 < Y \leq y_2) = F(x_2, y_2) - F(x_1, y_2) - F(x_2, y_1) + F(x_1, y_1) \geq 0.$$

(b) $F(x, y) \to 0$ as $x \to -\infty$ or $y \to -\infty$, and $F \to 1$ as both $\to +\infty$.
(c) F is right-continuous in each argument.

Proof. The rectangle identity is inclusion–exclusion for $\{X \leq x_2, Y \leq y_2\}$ minus the two strips. (b), (c) follow from continuity of probability (Theorem 1.2) as in one dimension. ∎

Example 2.1

Let $f(x, y) = c(x + y)$ on $0 \leq x, y \leq 1$. Then $\int_0^1 \int_0^1 (x + y)\, dy\, dx = 1$, so $c = 1$. The marginal is $f_1(x) = \int_0^1 (x + y)\, dy = x + \frac{1}{2}$ on $[0, 1]$, and by symmetry $f_2(y) = y + \frac{1}{2}$. Also $P(X + Y \leq 1) = \int_0^1 \int_0^{1-x} (x + y)\, dy\, dx = \int_0^1 \frac{1 - x^2}{2}\, dx = \frac{1}{3}$.

Example 2.2 (A discrete joint distribution)

An urn has 2 red, 3 blue and 4 green balls; draw 3 without replacement, and let X, Y be the numbers of red and blue balls. Then

$$f(x, y) = \binom{2}{x}\binom{3}{y}\binom{4}{3 - x - y} \Big/ \binom{9}{3}, \qquad x + y \leq 3,$$

e.g. $f(0, 0) = 4/84$ and $f(1, 1) = 24/84$. Summing out y (Vandermonde) gives the marginal $f_1(x) = \binom{2}{x}\binom{7}{3-x}/84$, i.e. $35, 42, 7$ over 84 for $x = 0, 1, 2$: the hypergeometric $H(3; 9, 2)$, as it must be.

Example 2.3 (Joint cdf)

For $f(x, y) = e^{-y}$ on $0 < x < y$: $F(x, y) = \int_0^{x \wedge y} \int_s^y e^{-u}\, du\, ds = \int_0^{x \wedge y} (e^{-s} - e^{-y})\, ds = 1 - e^{-x \wedge y} - (x \wedge y) e^{-y}$ for $x, y \geq 0$. Letting $y \to \infty$ gives $F_1(x) = 1 - e^{-x}$, and letting $x \to \infty$

gives $F_2(y) = 1 - e^{-y} - ye^{-y}$, the Gamma$(2,1)$ cdf.

2.2 Expectations, covariance and correlation

$E[g(X,Y)] = \sum\sum g(x,y)f(x,y)$ or $\int\int gf$; it is linear and monotone. If g depends only on x, $Eg(X)$ can be computed from the joint or from the marginal: integrating out y first gives $\int g(x)f_1(x)\,dx$.

Definition (Covariance and correlation)

$\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$ and, for positive variances, $\rho = \mathrm{Corr}(X,Y) = \mathrm{Cov}(X,Y)/(\sigma_1\sigma_2)$.

Theorem 2.2

(a) $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$ and $\mathrm{Cov}(X,X) = \mathrm{Var}\,X$.
(b) $\mathrm{Cov}(aX+b, cY+d) = ac\,\mathrm{Cov}(X,Y)$.
(c) $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$.

Proof. (a) is immediate.

(b) $aX + b - E(aX+b) = a(X-\mu_1)$, and similarly for Y.

(c) Expand $(X-\mu_1)(Y-\mu_2)$ and use linearity. ■

Theorem 2.3

(a) $\mathrm{Var}\left(\frac{Y-\mu_2}{\sigma_2} - \rho\frac{X-\mu_1}{\sigma_1}\right) = 1-\rho^2$.
(b) $-1 \le \rho \le 1$.
(c) $\rho = \pm 1$ iff $\frac{Y-\mu_2}{\sigma_2} = \pm\frac{X-\mu_1}{\sigma_1}$ with probability 1.

Proof. (a) Expand: $1 - 2\rho\cdot\rho + \rho^2 = 1-\rho^2$.

(b) A variance is nonnegative.

(c) $\rho = \pm 1$ makes the variance in (a) zero, so the variable is constant (zero, since its mean is zero) with probability 1 (Theorem 1.11). ■

Equivalently,

$$|\mathrm{Cov}(X,Y)| \le \sigma_1\sigma_2,$$

a case of the **Cauchy–Schwarz inequality**, $|E(VW)| \le \sqrt{EV^2\,EW^2}$. The correlation measures *linear* association, free of units.

Example 2.4 (Example 2.1 continued)

$EX = \int_0^1 x(x+\frac{1}{2})\,dx = \frac{7}{12}$, $EX^2 = \int_0^1 x^2(x+\frac{1}{2})\,dx = \frac{5}{12}$, so $\mathrm{Var}\,X = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$. Also $E(XY) = \int_0^1\int_0^1 xy(x+y)\,dy\,dx = \frac{1}{3}$, so $\mathrm{Cov}(X,Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$ and $\rho = -\frac{1}{11}$.

Joint moments and generating functions. The (r,s) joint moment is $m_{r,s} = E(X^rY^s)$. Expanding $e^{t_1X}e^{t_2Y} = \sum_{r,s}\frac{X^rY^s}{r!s!}t_1^rt_2^s$ suggests the **joint mgf** $M(t_1,t_2) = E(e^{t_1X+t_2Y})$, defined when finite for $|t_i| < h_i$.

Theorem 2.4 (Joint mgf)

If $M(t_1,t_2)$ exists near $(0,0)$:
(a) all joint moments exist, $E(X^rY^s)=\partial^{r+s}M/\partial t_1^r\partial t_2^s\big|_{\mathbf{0}}$, and $M(t_1,t_2)=\sum_{r,s}\frac{E(X^rY^s)}{r!s!}t_1^rt_2^s$;
(b) if two pairs have equal joint mgfs near $\mathbf{0}$, they have the same joint distribution.

Proof. (a) Repeat the proof of Theorem 1.7(a) with $e^{|t_1x|+|t_2y|}\le\sum_{\pm,\pm}e^{\pm t_1x\pm t_2y}$ as the dominating function.

(b) As in one dimension, this needs characteristic functions and is beyond these notes. ■

The marginal mgfs are $M_X(s)=M(s,0)$ and $M_Y(t)=M(0,t)$. The joint cumulant generating function $C(t_1,t_2)=\log M(t_1,t_2)=\sum\frac{c_{r,s}}{r!s!}t_1^rt_2^s$ has joint cumulants $c_{1,0}=EX$, $c_{0,1}=EY$, $c_{2,0}=\operatorname{Var}X$, $c_{1,1}=\operatorname{Cov}(X,Y)$, $c_{0,2}=\operatorname{Var}Y$. This follows by expanding $\log(1+A)$ as in one dimension.

Example 2.5

For $f(x,y)=e^{-y}$ on $0<x<y$ (Example 2.3),

$$M(t_1,t_2)=\int_0^\infty e^{t_1x}\int_x^\infty e^{-(1-t_2)y}\,dy\,dx=\frac{1}{(1-t_2)(1-t_1-t_2)},\qquad t_2<1,\ t_1+t_2<1.$$

Then $C=-\log(1-t_2)-\log(1-t_1-t_2)$, and differentiating at $\mathbf{0}$ gives $EX=1$, $EY=2$, $\operatorname{Var}X=1$, $\operatorname{Cov}(X,Y)=1$, $\operatorname{Var}Y=2$. So $\rho=1/\sqrt{2}$. Also $M(s,0)=(1-s)^{-1}$: $X\sim\text{Exp}(1)$.

2.3 Conditional distributions and conditional expectation

Motivation. Regression is about the distribution of Y given X. The conditional expectation $E[Y\mid X]$ is the best predictor of Y from X, and the identities $E[E[Y\mid X]]=E[Y]$ and the variance decomposition let us compute moments by conditioning.

Definition (Conditional pmf/pdf)

For $f_1(x)>0$, $f_{2|1}(y\mid x)=f(x,y)/f_1(x)$. For each fixed x it is a pmf/pdf in y.

In the continuous case $P(X=x)=0$, so this is justified as a limit: if f and f_1 are continuous,

$$P(c\le Y\le d\mid x\le X\le x+h)=\frac{\frac1h\int_x^{x+h}\int_c^d f(s,y)\,dy\,ds}{\frac1h\int_x^{x+h}f_1(s)\,ds}\xrightarrow{h\downarrow0}\int_c^d f_{2|1}(y\mid x)\,dy.$$

Theorem 2.5

$P(a\le X\le b,c\le Y\le d)=\int_a^b P(c\le Y\le d\mid X=x)f_1(x)\,dx$ (a sum in the discrete case).

Proof. Substitute $f(x,y)=f_{2|1}(y\mid x)f_1(x)$ into the double integral and integrate in y first. ■

Example 2.6 (Example 2.2 continued)

Given no red ball ($X=0$), $f_{2|1}(y\mid 0)=f(0,y)/f_1(0)=\binom{3}{y}\binom{4}{3-y}/35$, i.e. $4,18,12,1$ over 35 for $y=0,1,2,3$. This is $H(3;7,3)$: once the red balls are excluded, we are effectively drawing from the 7 non-red balls.

Example 2.7

For $f(x,y) = e^{-y}$ on $0 < x < y$, $f_{2|1}(y \mid x) = e^{-(y-x)}$ for $y > x$: given $X = x$, $Y - x \sim \text{Exp}(1)$. Hence $P(Y > c \mid X = x) = e^{-(c \vee x - x)}$ and, by Theorem 2.5, $P(X \le 1, Y > 2) = \int_0^1 e^{-(2-x)} e^{-x}\, dx = e^{-2}$.

Definition (Conditional expectation and variance)

$E[g(X,Y) \mid X = x] = \int g(x,y) f_{2|1}(y \mid x)\, dy$; $\mu_{2|1}(x) = E(Y \mid X = x)$; $\text{Var}(Y \mid X = x) = E[(Y - \mu_{2|1}(x))^2 \mid X = x] = E(Y^2 \mid X = x) - \mu_{2|1}(x)^2$. Replacing x by X gives the random variables $E(Y \mid X)$ and $\text{Var}(Y \mid X)$.

Conditional expectation is linear and monotone, and functions of the conditioning variable factor out: $E[c(X)g(Y) \mid X = x] = c(x)E[g(Y) \mid X = x]$.

Theorem 2.6

(a) (Tower property) $E[E(Y \mid X)] = E(Y)$.
(b) $\text{Cov}(Y - E(Y \mid X), v(X)) = 0$ for every function v with $Ev(X)^2 < \infty$.

Proof. (a) $\int \left(\int y f_{2|1}(y \mid x)\, dy \right) f_1(x)\, dx = \iint y f(x,y)\, dy\, dx = EY$.

(b) Let $Z = Y - E(Y \mid X)$. Then $E(Z \mid X) = 0$, so $EZ = 0$ by (a), and $E[Zv(X)] = E[v(X)E(Z \mid X)] = 0$. ■

Theorem 2.7 (Best predictor)

Among all functions u, $E[(Y - u(X))^2]$ is minimized by $u(X) = E(Y \mid X)$, and the minimum is $E[\text{Var}(Y \mid X)]$.

Proof. Write $Y - u(X) = (Y - E(Y \mid X)) + v(X)$ with $v = E(Y \mid X) - u(X)$. The cross term has expectation 0 by Theorem 2.6(b), so $E(Y-u)^2 = E(Y - E(Y \mid X))^2 + Ev(X)^2$, minimized when $v = 0$. And $E(Y - E(Y \mid X))^2 = E[E((Y - E(Y \mid X))^2 \mid X)] = E\,\text{Var}(Y \mid X)$. ■

$E(Y \mid X)$ is called the **regression function** of Y on X.

Theorem 2.8 (Variance decomposition)

$\text{Var}\, Y = E[\text{Var}(Y \mid X)] + \text{Var}[E(Y \mid X)]$.

Proof. $Y - \mu = (Y - E(Y \mid X)) + (E(Y \mid X) - \mu)$; the two parts are uncorrelated by Theorem 2.6(b). Square and take expectations, using $E[E(Y \mid X)] = \mu$ and Theorem 2.7. ■

Geometrically $Y - EY$ splits into orthogonal pieces: the part explained by X and the residual. This is the basis of regression and ANOVA (Chapter 10).

Example 2.8 (Example 2.1 continued)

$f_{2|1}(y \mid x) = \frac{x+y}{x+1/2}$ on $[0,1]$, so $E(Y \mid X = x) = \frac{x/2+1/3}{x+1/2} = \frac{3x+2}{6x+3}$. The regression is not linear, and it decreases in x, consistent with $\rho < 0$.

Example 2.9 (Checking the decomposition)

In Example 2.7, $E(Y \mid X) = X+1$ and $\operatorname{Var}(Y \mid X) = 1$. So $E[E(Y \mid X)] = EX+1 = 2 = EY$, and $E\operatorname{Var}(Y \mid X) + \operatorname{Var} E(Y \mid X) = 1 + \operatorname{Var} X = 2 = \operatorname{Var} Y$, in agreement with Example 2.5. Also $\operatorname{Cov}(Y - E(Y \mid X), X) = \operatorname{Cov}(Y, X) - \operatorname{Var} X = 0$, as Theorem 2.6(b) requires.

2.4 Independence

Motivation. Random samples are independent observations, and independence is what makes the joint density factor and the variance of a sum equal the sum of variances.

Definition (Independent random variables)

X and Y are independent if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all (Borel) sets A, B.

Theorem 2.9

Each of the following is equivalent to independence:
(a) $F(x, y) = F_1(x)F_2(y)$ for all x, y;
(b) $f(x, y) = f_1(x)f_2(y)$ for all x, y;
(c) $M(t_1, t_2) = M_X(t_1)M_Y(t_2)$ near 0 (when the mgf exists). Moreover (b) holds iff $f(x, y) = g_1(x)g_2(y)$ for *some* functions g_1, g_2.

Proof. Rectangles determine the distribution, so (a)$\Leftrightarrow$ independence.

(a) and (b) are equivalent by differentiation and integration.

(c) follows from the uniqueness of the mgf. If $f = g_1g_2$, integrating gives $f_1 = g_1 \int g_2$, $f_2 = g_2 \int g_1$ and $1 = \int g_1 \int g_2$, so $f = f_1f_2$. ■

The factorization must hold on the whole plane, *support included*: $f(x, y) = 8xy$ on $0 \le x \le y \le 1$ looks like a product, but the triangular support makes X, Y dependent.

Theorem 2.10

If X, Y are independent then (a) $g_1(X)$ and $g_2(Y)$ are independent;
(b) $E[g_1(X)g_2(Y)] = Eg_1(X)\, Eg_2(Y)$;
(c) $\operatorname{Cov}(X, Y) = 0$;
(d) $\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y$. In general $\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X, Y)$.

Proof. (a) $\{g_1(X) \in A\} = \{X \in g_1^{-1}(A)\}$.

(b) Factor the double integral.

(c) is (b) with identities and Theorem 2.2(c).

(d) Expand the square. ■

Uncorrelated does not imply independent: if X is uniform on $\{-1, 0, 1\}$ and $Y = X^2$, then $\operatorname{Cov}(X, Y) = EX^3 = 0$, yet Y is a function of X.

Example 2.10

(a) $f(x,y) = 6e^{-2x-3y}$ on $x,y>0$ factors as $2e^{-2x}\cdot 3e^{-3y}$ on a product set, so $X\sim\operatorname{Exp}(\frac{1}{2})$ and $Y\sim\operatorname{Exp}(\frac{1}{3})$ are independent.
(b) $f(x,y)=e^{-y}$ on $0<x<y$ does not factor ($\rho=1/\sqrt{2}$).
(c) $f(x,y)=\frac{9}{40}(1+x^2y^2)$ on $[-1,1]^2$ (the constant makes it integrate to 1). By the symmetries $(x,y)\mapsto(-x,y)$ and $(x,-y)$, $EX=EY=E(XY)=0$, so $\operatorname{Cov}(X,Y)=0$. But $f_1(x)=\frac{9}{20}(1+\frac{x^2}{3})$ and likewise f_2, so $f(1,1)=\frac{9}{20}\neq f_1(1)f_2(1)=\frac{9}{25}$: X and Y are dependent.

2.5 Random vectors

For $\mathbf{X}=(X_1,\dots,X_k)'$ the joint pmf/pdf, marginals (sum or integrate out the other coordinates), cdf and expectations are defined as above. The **mean vector** is $E\mathbf{X}=(EX_1,\dots,EX_k)'$, and the **covariance matrix** is

$$\operatorname{Var}(\mathbf{X})=E[(\mathbf{X}-\mu)(\mathbf{X}-\mu)']=\big(\operatorname{Cov}(X_i,X_j)\big)_{i,j}.$$

More generally $\operatorname{Cov}(\mathbf{X},\mathbf{Y})=E[(\mathbf{X}-\mu)(\mathbf{Y}-\eta)']$.

Theorem 2.11

For constant matrices A,C and vectors $\mathbf{b},\mathbf{d}$:

(a) $E(A\mathbf{X}+\mathbf{b})=AE\mathbf{X}+\mathbf{b}$;
(b) $\operatorname{Var}(A\mathbf{X}+\mathbf{b})=A\operatorname{Var}(\mathbf{X})A'$ and $\operatorname{Cov}(A\mathbf{X}+\mathbf{b},C\mathbf{Y}+\mathbf{d})=A\operatorname{Cov}(\mathbf{X},\mathbf{Y})C'$;
(c) $\operatorname{Var}(\mathbf{X}+\mathbf{Y})=\operatorname{Var}\mathbf{X}+\operatorname{Var}\mathbf{Y}+\operatorname{Cov}(\mathbf{X},\mathbf{Y})+\operatorname{Cov}(\mathbf{Y},\mathbf{X})$;
(d) $\operatorname{Var}(\mathbf{X})$ is symmetric and nonnegative definite.

Proof. (a)–(c) follow from $E(CWD)=CE(W)D$ for a random matrix W, applied to $(\mathbf{X}-\mu)(\mathbf{Y}-\eta)'$.

(d) $\mathbf{a}'\operatorname{Var}(\mathbf{X})\mathbf{a}=\operatorname{Var}(\mathbf{a}'\mathbf{X})\geq 0$. ■

By the spectral theorem $\operatorname{Var}\mathbf{X}=P\operatorname{diag}(\lambda_1,\dots,\lambda_k)P'$ with P orthogonal and $\lambda_1\geq\cdots\geq\lambda_k\geq 0$. Then $P'\mathbf{X}$ has uncorrelated coordinates with variances λ_i: these are the **principal components**, and the eigenvectors give the main directions of spread.

Generating functions. $M(\mathbf{t})=E(e^{\mathbf{t}'\mathbf{X}})$ yields all joint moments and determines the distribution. The cgf $C=\log M$ gives $E\mathbf{X}=\nabla C(\mathbf{0})$ and $\operatorname{Var}\mathbf{X}=\nabla^2 C(\mathbf{0})$ (gradient and Hessian).

Conditioning. $f_{\mathbf{Y}|\mathbf{X}}(\mathbf{y}\mid\mathbf{x})=f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})/f_{\mathbf{X}}(\mathbf{x})$. The tower property, the orthogonality $\operatorname{Cov}(\mathbf{Y}-E(\mathbf{Y}\mid\mathbf{X}),v(\mathbf{X}))=0$, the matrix decomposition

$$\operatorname{Var}\mathbf{Y}=E[\operatorname{Var}(\mathbf{Y}\mid\mathbf{X})]+\operatorname{Var}[E(\mathbf{Y}\mid\mathbf{X})],$$

and the best-predictor property for $E\|\mathbf{Y}-u(\mathbf{X})\|^2$ carry over with the same proofs.

Independence. $X_1,\dots,X_n$ are independent iff the joint pdf (or the mgf) factors into marginals, again iff it factors into *some* functions of the separate arguments. Then functions of separate components are independent, expectations of products factor, the covariance matrix of $\sum_i\mathbf{X}_i$ is $\sum_i\operatorname{Var}\mathbf{X}_i$, and $M_{\sum\mathbf{X}_i}(\mathbf{t})=\prod_i M_{\mathbf{X}_i}(\mathbf{t})$.

Example 2.11

Let X_1, X_2, X_3 be i.i.d. standard exponential and $Y_1 = X_1$, $Y_2 = X_1 + X_2$. Then $\operatorname{Var}(Y_1, Y_2)' = A\, I\, A'$ with $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, i.e. $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, so $\operatorname{Corr}(Y_1, Y_2) = 1/\sqrt{2}$.

2.6 Selected exercises

Motivation. These are the course's exercises for Chapter 2 (numbered as there). They use conditioning to find a marginal law, linear algebra to maximize a correlation, and an explicit density to separate pairwise from mutual independence.

Exercise 2.7 (Pólya's urn)

An urn contains $b \geq 0$ black and $r \geq 0$ red balls, $b + r > 0$. A ball is drawn; set $X_1 = 1$ if it is black and $X_1 = 0$ if red, and return it together with $c \geq 0$ further balls of the same colour. This is repeated, giving $X_2, X_3, \ldots$.
(a) Find the conditional distribution of X_{n+1} given $X_1, \ldots, X_n$.
(b) Find the marginal distribution of X_n, $n \geq 1$.

Solution.

(a) Let $S_n = X_1 + \cdots + X_n$ and $N_n = b + r + nc$. After n draws the urn holds N_n balls, of which $b + cS_n$ are black. The next ball is drawn at random, so

$$P(X_{n+1} = 1 \mid X_1, \ldots, X_n) = \frac{b + cS_n}{N_n} =: M_n, \qquad P(X_{n+1} = 0 \mid X_1, \ldots, X_n) = 1 - M_n.$$

(b)

Step 1. M_n is the proportion of black balls after n draws. If $X_{n+1} = 1$ the urn then holds $M_n N_n + c$ black balls among $N_n + c$; otherwise $M_n N_n$. By (a),

$$E[M_{n+1} \mid X_1, \ldots, X_n] = M_n \frac{M_n N_n + c}{N_n + c} + (1 - M_n) \frac{M_n N_n}{N_n + c} = \frac{M_n (N_n + c)}{N_n + c} = M_n.$$

Step 2. Taking expectations (Theorem 2.6), $EM_{n+1} = EM_n = \cdots = EM_0 = b/(b + r)$.

Step 3. Hence $P(X_{n+1} = 1) = E\big[P(X_{n+1} = 1 \mid X_1, \ldots, X_n)\big] = EM_n = b/(b + r)$. So every X_n is Bernoulli$(b/(b + r))$, as for the first draw, although the X_n are dependent when $c > 0$. ∎

Exercise 2.12 (Maximal correlation of linear combinations)

Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be random vectors of dimensions k_1 and k_2 with $\Sigma_{11} = \operatorname{Var} \mathbf{X}_1$ and $\Sigma_{22} = \operatorname{Var} \mathbf{X}_2$ positive definite and $\Sigma_{12} = \operatorname{Cov}(\mathbf{X}_1, \mathbf{X}_2) = \Sigma_{21}'$. Find nonzero $\mathbf{a}_1, \mathbf{a}_2$ maximizing $\operatorname{Corr}(\mathbf{a}_1' \mathbf{X}_1, \mathbf{a}_2' \mathbf{X}_2)$, and the maximum.

Solution.

Step 1. By Theorem 2.11,

$$\operatorname{Corr}(\mathbf{a}_1' \mathbf{X}_1, \mathbf{a}_2' \mathbf{X}_2) = \frac{\mathbf{a}_1' \Sigma_{12} \mathbf{a}_2}{\sqrt{\mathbf{a}_1' \Sigma_{11} \mathbf{a}_1} \sqrt{\mathbf{a}_2' \Sigma_{22} \mathbf{a}_2}}.$$

Put $\mathbf{u} = \Sigma_{11}^{1/2}\mathbf{a}_1$, $\mathbf{v} = \Sigma_{22}^{1/2}\mathbf{a}_2$ (symmetric positive definite square roots) and $K = \Sigma_{11}^{-1/2}\Sigma_{12}\Sigma_{22}^{-1/2}$. As $\mathbf{a}_1, \mathbf{a}_2$ range over nonzero vectors, so do $\mathbf{u}, \mathbf{v}$, and the correlation equals $\mathbf{u}'K\mathbf{v}/(\|\mathbf{u}\|\|\mathbf{v}\|)$.

Step 2. Let $\lambda_1 \geq 0$ be the largest eigenvalue of the symmetric nonnegative definite matrix $K'K$, with unit eigenvector $\mathbf{e}_1$. By the spectral theorem $\|K\mathbf{v}\|^2 = \mathbf{v}'K'K\mathbf{v} \leq \lambda_1\|\mathbf{v}\|^2$, and by the Cauchy–Schwarz inequality

$$\mathbf{u}'K\mathbf{v} \leq \|\mathbf{u}\|\,\|K\mathbf{v}\| \leq \sqrt{\lambda_1}\,\|\mathbf{u}\|\,\|\mathbf{v}\|.$$

So the correlation is at most $\sqrt{\lambda_1}$.

Step 3. If $\lambda_1 > 0$, take $\mathbf{v} = \mathbf{e}_1$ and $\mathbf{u} = K\mathbf{e}_1$. Then $\|\mathbf{u}\|^2 = \mathbf{e}_1'K'K\mathbf{e}_1 = \lambda_1$ and $\mathbf{u}'K\mathbf{v} = \lambda_1$, so the correlation equals $\lambda_1/\sqrt{\lambda_1} = \sqrt{\lambda_1}$. If $\lambda_1 = 0$, then $K = 0$, i.e. $\Sigma_{12} = 0$, and every correlation is 0.

Step 4. Translating back, the maximum is $\sqrt{\lambda_1}$, attained at

$$\mathbf{a}_2 = \Sigma_{22}^{-1/2}\mathbf{e}_1, \qquad \mathbf{a}_1 = \Sigma_{11}^{-1/2}K\mathbf{e}_1 = \Sigma_{11}^{-1}\Sigma_{12}\mathbf{a}_2$$

(and at their positive multiples). Since $K'K = \Sigma_{22}^{-1/2}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22}^{-1/2}$ is similar to $\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}$, λ_1 is also the largest eigenvalue of that matrix. The maximum $\sqrt{\lambda_1}$ is the first *canonical correlation*; for $k_1 = k_2 = 1$ it is $|\mathrm{Corr}(X_1, X_2)|$. ■

Exercise 2.15 (Pairwise but not mutually independent)

Let (X, Y, Z) have density

$$f(x, y, z) = \frac{1 - \sin x \sin y \sin z}{8\pi^3}, \qquad 0 \leq x, y, z \leq 2\pi,$$

and 0 otherwise. Show that any two of X, Y, Z are independent, but X, Y, Z are not mutually independent.

Solution.

Step 1. $f \geq 0$ because $|\sin x \sin y \sin z| \leq 1$, and $\int_0^{2\pi} \sin t\, dt = 0$ gives $\iiint f = (2\pi)^3/(8\pi^3) = 1$, so f is a density.

Step 2. Integrating out z and using $\int_0^{2\pi} \sin z\, dz = 0$, the density of (X, Y) is $f_{XY}(x, y) = 2\pi/(8\pi^3) = 1/(4\pi^2)$ on $[0, 2\pi]^2$. Integrating once more, X and Y are uniform on $[0, 2\pi]$. Thus $f_{XY} = f_X f_Y$, and X, Y are independent by Theorem 2.9(b). By symmetry of f in (x, y, z) the same holds for (X, Z) and (Y, Z).

Step 3. Choose $\delta \in (0, \pi/2)$ so small that $\sin t \geq 2^{-1/3}$ for $t \in I = [\pi/2 - \delta, \pi/2 + \delta]$. On the box I^3 we then have $\sin x \sin y \sin z \geq \frac{1}{2}$, so $f \leq \frac{1}{2} \cdot \frac{1}{8\pi^3}$ there and

$$P(X \in I, Y \in I, Z \in I) \leq \frac{1}{2} \cdot \frac{(2\delta)^3}{8\pi^3} < \left(\frac{2\delta}{2\pi}\right)^3 = P(X \in I)P(Y \in I)P(Z \in I).$$

So the product rule fails for the events $\{X \in I\}, \{Y \in I\}, \{Z \in I\}$, and X, Y, Z are not mutually independent. ■

Chapter 3. Common Distributions

Conventions

Many distributions below are introduced through a **representation**: $X \overset{d}{=} Z_1 + \cdots + Z_n$ means X has the same distribution as a sum of the indicated variables. A representation makes means, variances and mgfs almost free (Theorem 2.10 and the product rule for mgfs of independent sums). Throughout, $\mathrm{Exp}(\beta)$ and $\mathrm{Gamma}(\alpha, \beta)$ use the *scale* β, so their means are β and $\alpha\beta$.

3.1 The hypergeometric distribution

A population of N units contains D of type "1" and $N-D$ of type "0". The population proportion is $p = D/N$. Draw n units by **simple random sampling**: one at a time, each remaining unit equally likely, without replacement. The number X of type-1 units in the sample has the **hypergeometric distribution** $X \sim H(n; N, D)$:

$$f(x) = \binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n}, \qquad 0 \le x \le D,\ 0 \le n-x \le N-D.$$

Proposition 3.1

$$EX = np \text{ and } \operatorname{Var} X = \frac{N-n}{N-1}\, np(1-p).$$

Proof. Using $x\binom{D}{x} = D\binom{D-1}{x-1}$,

$$EX = D\sum_x \binom{D-1}{x-1}\binom{N-1-(D-1)}{n-1-(x-1)}\Big/\binom{N}{n} = D\binom{N-1}{n-1}\Big/\binom{N}{n} = \frac{nD}{N}.$$

The middle sum is Vandermonde's identity, obtained by comparing the coefficient of t^{n-1} on both sides of $(1+t)^{D-1}(1+t)^{N-D} = (1+t)^{N-1}$. In the same way $x(x-1)\binom{D}{x} = D(D-1)\binom{D-2}{x-2}$ gives $E[X(X-1)] = n(n-1)\frac{D(D-1)}{N(N-1)}$. Then

$$\operatorname{Var} X = E[X(X-1)] + EX - (EX)^2 = \frac{N-n}{N-1}\, n\frac{D}{N}\Big(1 - \frac{D}{N}\Big).$$

■

Example 3.1 (Acceptance sampling)

A lot of $N = 40$ items contains $D = 4$ defectives. We inspect $n = 5$ at random and accept the lot if at most c defectives are found. Then

$$P(X=0) = \binom{36}{5}\Big/\binom{40}{5} = \frac{376992}{658008} \approx 0.573, \qquad P(X=1) = \frac{4\binom{36}{4}}{\binom{40}{5}} = \frac{235620}{658008} \approx 0.358.$$

So $P(X \le 1) \approx 0.931$ and $P(X \le 2) \approx 0.996$. The rule "accept if $X \le 2$" accepts such a lot more than 99% of the time, so to reject a lot with 10% defectives with probability at least 0.99 we would need a smaller acceptance number or a larger sample.

Binomial approximation. If $N \to \infty$ with $D/N = p$ fixed and n fixed, then

$$\binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n} = \binom{n}{x}\frac{D(D-1)\cdots(D-x+1)\cdot(N-D)\cdots(N-D-n+x+1)}{N(N-1)\cdots(N-n+1)}$$
$$\to \binom{n}{x}p^x(1-p)^{n-x}.$$

Drawing without replacement from a huge population is practically the same as drawing with replacement.

3.2 Binomial and multinomial distributions

If each draw is made *with* replacement, the outcomes $Z_1, \ldots, Z_n$ are independent with $P(Z_i = 1) = p$: **Bernoulli trials**, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. The number of successes is binomial.

Definition (Binomial)

$X \sim B(n,p)$ if $f(x) = \binom{n}{x}p^x(1-p)^{n-x}$, $x = 0, \ldots, n$. Representation: $X \overset{d}{=} Z_1 + \cdots + Z_n$, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$.

Proposition 3.2

(a) $EX = np$, $\operatorname{Var} X = np(1-p)$.
(b) $M(t) = (pe^t + q)^n$ with $q = 1-p$.
(c) If $X_1 \sim B(n_1, p)$ and $X_2 \sim B(n_2, p)$ are independent, then $X_1 + X_2 \sim B(n_1 + n_2, p)$.

Proof. (a) Directly, $EX = \sum x\binom{n}{x}p^xq^{n-x} = np\sum_k \binom{n-1}{k}p^kq^{n-1-k} = np$, and similarly $E[X(X-1)] = n(n-1)p^2$; so $\operatorname{Var} X = n(n-1)p^2 + np - n^2p^2 = np(1-p)$. (Or add the Bernoulli means p and variances $p(1-p)$.) (b) A Bernoulli has mgf $pe^t + q$, and mgfs of independent summands multiply.

(c) The product of the mgfs is $(pe^t + q)^{n_1+n_2}$; uniqueness of mgfs (Theorem 1.7) identifies the distribution. ■

Example 3.2 (Binomial conditioned on its total is hypergeometric)

Let $X_1, \ldots, X_N \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $D \le N$. Then the conditional distribution of $X_1 + \cdots + X_D$ given $X_1 + \cdots + X_N = n$ is $H(n; N, D)$. Indeed $X_1 + \cdots + X_D \sim B(D,p)$ and $X_{D+1} + \cdots + X_N \sim B(N-D,p)$ are independent, so

$$P\Big(\sum_{i\le D} X_i = x \,\Big|\, \sum_{i\le N} X_i = n\Big) = \frac{\binom{D}{x}p^xq^{D-x}\binom{N-D}{n-x}p^{n-x}q^{N-D-n+x}}{\binom{N}{n}p^nq^{N-n}} = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}.$$

The parameter p cancels.

Definition (Multinomial)

Each of n independent trials falls in category j with probability p_j, $\sum_j p_j = 1$. The counts $\mathbf{X} = (X_1, \ldots, X_k)$ satisfy

$$f(\mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum_j x_j = n,$$

written $\mathbf{X} \sim \text{Multi}(n, \mathbf{p})$. Representation: $\mathbf{X} \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with $\mathbf{Z}_i \overset{\text{i.i.d.}}{\sim} \text{Multi}(1, \mathbf{p})$ (one-hot vectors). For $k = 2$, $(X, n - X)$ with $X \sim B(n, p_1)$.

Proposition 3.3

(a) $EX_l = np_l$, $\operatorname{Var} X_l = np_l(1 - p_l)$, $\operatorname{Cov}(X_l, X_m) = -np_lp_m$ for $l \neq m$; i.e. $\operatorname{Var} \mathbf{X} = n\{\operatorname{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}'\}$.
(b) $M(\mathbf{t}) = (p_1 e^{t_1} + \cdots + p_k e^{t_k})^n$.

Proof. For one trial, $Z_{1l} \in \{0, 1\}$ with $P(Z_{1l} = 1) = p_l$, and $Z_{1l}Z_{1m} = 0$ for $l \neq m$. So $\operatorname{Cov}(Z_{1l}, Z_{1m}) = -p_lp_m$, and $M_{\mathbf{Z}_1}(\mathbf{t}) = \sum_l p_l e^{t_l}$. Sum over n i.i.d. trials (Theorem 2.11). ■

The negative covariances express the constraint $\sum_j X_j = n$.

3.3 Geometric and negative binomial distributions

In Bernoulli trials with success probability p, the trial number W_1 of the first success satisfies $P(W_1 = x) = q^{x-1}p$, $x = 1, 2, \ldots$: the **geometric distribution** $\text{Geo}(p)$.

Proposition 3.4

(a) $M_{W_1}(t) = pe^t(1 - qe^t)^{-1}$ for $t < -\log q$.
(b) $EW_1 = 1/p$, $\operatorname{Var} W_1 = q/p^2$.
(c) (Memorylessness) $P(W_1 > m + k \mid W_1 > m) = P(W_1 > k)$.

Proof. (a) A geometric series: $\sum_x e^{tx} q^{x-1} p = pe^t \sum_x (qe^t)^{x-1}$.

(b) The cgf is $C(t) = t + \log p - \log(1 - qe^t)$. Expanding $-\log(1 - qe^t)$ in powers of t gives $C(t) = \frac{1}{p}t + \frac{q}{p^2}\frac{t^2}{2} + \cdots$.

(c) $P(W_1 > k) = q^k$. ■

The trial number W_r of the rth success needs $r - 1$ successes in the first $x - 1$ trials and a success at trial x:

$$P(W_r = x) = \binom{x-1}{r-1} p^r q^{x-r}, \qquad x = r, r+1, \ldots.$$

This is the **negative binomial distribution** $\text{Negbin}(r, p)$. The name comes from the expansion $(1 - q)^{-r} = \sum_{k \geq 0} \binom{r+k-1}{k} q^k$; substituting $x = r + k$ shows that the probabilities sum to $p^r p^{-r} = 1$. The waiting times $W_1, W_2 - W_1, \ldots, W_r - W_{r-1}$ are i.i.d. $\text{Geo}(p)$, since

$$P(W_1 = x_1, \ldots, W_r - W_{r-1} = x_r) = \prod_{i=1}^{r} q^{x_i - 1} p.$$

Hence the representation $\text{Negbin}(r, p) \overset{d}{=} \text{Geo}_1(p) \oplus \cdots \oplus \text{Geo}_r(p)$, where $\oplus$ denotes a sum of independent variables.

Proposition 3.5

For $X \sim \text{Negbin}(r,p)$: $EX = r/p$, $\text{Var}\, X = rq/p^2$, $M(t) = \{pe^t(1-qe^t)^{-1}\}^r$. Independent $\text{Negbin}(r_i,p)$ variables sum to $\text{Negbin}(\sum r_i, p)$.

Proof. Sum r i.i.d. geometrics (Proposition 3.4); multiply mgfs and use uniqueness. ■

3.4 The Poisson distribution

Motivation. Counts of rare events are modelled by the Poisson distribution, the limit of binomials with np fixed. Its additivity and the link with exponential waiting times make it the basic count model.

Proposition 3.6 (Poisson approximation of the binomial)

If $n \to \infty$ and $np_n \to \lambda > 0$, then $\binom{n}{x} p_n^x (1-p_n)^{n-x} \to e^{-\lambda}\lambda^x/x!$.

Proof.

$$\binom{n}{x} p_n^x (1-p_n)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{n^x} \cdot \frac{(np_n)^x}{x!} \cdot \Big(1 - \frac{np_n}{n}\Big)^n (1-p_n)^{-x}.$$

The first factor tends to 1, the second to $\lambda^x/x!$, the third to $e^{-\lambda}$ (since $(1+a_n/n)^n \to e^a$ when $a_n \to a$), and the last to 1. ■

$X \sim \text{Poisson}(\lambda)$ if $f(x) = e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \ldots$.

Example 3.3

A batter hits a home run in 2% of at-bats. Over 300 at-bats, $X \sim B(300, 0.02) \approx \text{Poisson}(6)$. Then $P(X = 10) \approx e^{-6}6^{10}/10! \approx 0.041$; the exact binomial value is 0.040.

Proposition 3.7

(a) $M(t) = \exp\{\lambda(e^t - 1)\}$ for all t.
(b) $EX = \text{Var}\, X = \lambda$.
(c) Independent $\text{Poisson}(\lambda_i)$ variables sum to $\text{Poisson}(\sum \lambda_i)$.

Proof. (a) $\sum_x e^{tx} e^{-\lambda}\lambda^x/x! = e^{-\lambda}e^{\lambda e^t}$.

(b) The cgf $\lambda(e^t - 1) = \lambda \sum_{r \geq 1} t^r/r!$ has every cumulant equal to λ.

(c) Multiply mgfs and use uniqueness. ■

The Poisson process. Let N_t count the events in $[0,t]$, with

(a) (stationarity) $N_{s+t} - N_s \overset{d}{=} N_t$ and $N_0 = 0$;
(b) (independent increments) counts on disjoint intervals are independent;
(c) (proportionality) $P(N_h = 1) = \lambda h + o(h)$;
(d) (rareness) $P(N_h \geq 2) = o(h)$ as $h \downarrow 0$.

Theorem 3.8

Under (a)–(d), $N_t \sim \text{Poisson}(\lambda t)$.

Proof. Let $g(x,t) = P(N_t = x)$. Splitting according to the increment on $(t, t+h]$ and using (a), (b), (d),

$$g(x,t+h) = g(x,t)P(N_h = 0) + g(x-1,t)P(N_h = 1) + o(h) = g(x,t)(1-\lambda h) + g(x-1,t)\lambda h + o(h).$$

Hence $\partial_t g(x,t) = -\lambda g(x,t) + \lambda g(x-1,t)$, i.e. $\frac{d}{dt}\{e^{\lambda t} g(x,t)\} = \lambda e^{\lambda t} g(x-1,t)$, with $g(0,0) = 1$ and $g(x,0) = 0$ for $x \geq 1$. For $x = 0$ this gives $g(0,t) = e^{-\lambda t}$. Inductively, if $g(x-1,t) = e^{-\lambda t}(\lambda t)^{x-1}/(x-1)!$, then $e^{\lambda t} g(x,t) = \int_0^t \lambda(\lambda s)^{x-1}/(x-1)!\, ds = (\lambda t)^x/x!$. ∎

Heuristically: split $[0,t]$ into n pieces; each contains an event with probability about $\lambda t/n$, independently, so $N_t \approx B(n, \lambda t/n) \to \text{Poisson}(\lambda t)$.

Example 3.4

A cable has on average 0.04 flaws per meter. In 150 meters the number of flaws is Poisson(6), so $P(\text{at least } 10) = 1 - \sum_{x \leq 9} e^{-6} 6^x/x! \approx 0.084$.

3.5 Exponential and gamma distributions

In a Poisson process with rate λ, the waiting time W_1 for the first event satisfies $\{W_1 > t\} = \{N_t = 0\}$. So $P(W_1 \leq t) = 1 - e^{-\lambda t}$, and W_1 has density $\lambda e^{-\lambda t}$, $t > 0$: the **exponential distribution** $\text{Exp}(1/\lambda)$, with mean $1/\lambda$.

Proposition 3.9

$M_{W_1}(t) = (1 - t/\lambda)^{-1}$ for $t < \lambda$; $EW_1 = 1/\lambda$, $\operatorname{Var} W_1 = 1/\lambda^2$; and $P(W_1 > s + t \mid W_1 > s) = P(W_1 > t)$.

Proof. $\int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda/(\lambda - t)$. The cgf $-\log(1 - t/\lambda) = t/\lambda + \frac{1}{2}(t/\lambda)^2 + \cdots$ gives the mean and variance. Memorylessness: $e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. ∎

For the rth event, $\{W_r > t\} = \{N_t \leq r-1\}$, so $P(W_r \leq t) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t}(\lambda t)^k/k!$. Differentiating, the sum telescopes:

$$f_{W_r}(t) = \lambda e^{-\lambda t}\Big\{\sum_{k=0}^{r-1} \frac{(\lambda t)^k}{k!} - \sum_{k=1}^{r-1} \frac{(\lambda t)^{k-1}}{(k-1)!}\Big\} = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}, \qquad t > 0.$$

Allowing any shape $\alpha > 0$ gives the gamma family.

Definition (Gamma)

$X \sim \text{Gamma}(\alpha, \beta)$ if $f(x) = \dfrac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$. Here α is the shape and β the scale, and $\text{Exp}(\beta) = \text{Gamma}(1, \beta)$.

Integration by parts gives $\Gamma(\alpha) = (\alpha-1)\Gamma(\alpha-1)$, so $\Gamma(n) = (n-1)!$. Also $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (substitute $x = z^2/2$ and use $\int \phi = 1$).

Proposition 3.10

(a) $EX = \alpha\beta$, $\operatorname{Var} X = \alpha\beta^2$.
(b) $M(t) = (1-\beta t)^{-\alpha}$ for $t < 1/\beta$.
(c) Independent Gamma(α_i, β) variables with a common scale sum to Gamma$(\sum \alpha_i, \beta)$. In particular Gamma$(r, \beta) \stackrel{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \overset{\text{i.i.d.}}{\sim}$ Exp(β).

Proof. (a) Substituting $y = x/\beta$, $EX^k = \beta^k \Gamma(\alpha+k)/\Gamma(\alpha)$, which gives $\alpha\beta$ and $\alpha(\alpha+1)\beta^2$.

(b) Substitute $y = (1/\beta - t)x$ in $\int e^{tx} f(x)\,dx$.

(c) Multiply the mgfs. ■

In a Poisson process the inter-event times $W_1, W_2 - W_1, \ldots$ are i.i.d. Exp$(1/\lambda)$, and W_r is their sum.

3.6 The normal distribution

Motivation. The normal distribution appears as the limit of sums (central limit theorem) and has unusually convenient algebra: linear combinations of independent normals are normal. Most classical inference is built on it.

Definition (Normal)

$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density, with cdf Φ. $X \sim N(\mu, \sigma^2)$ has density $\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$.

$\int \phi = 1$: $(\int e^{-z^2/2}\,dz)^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$.

Proposition 3.11

(a) $EX = \mu$, $\operatorname{Var} X = \sigma^2$.
(b) $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$ for all t.
(c) If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof. (a) With $z = (x-\mu)/\sigma$: $EX = \mu + \sigma \int z\phi(z)\,dz = \mu$ by symmetry. Also $\int z^2\phi = 2\int_0^\infty z^2\phi(z)\,dz = \frac{2}{\sqrt{\pi}}\Gamma(\frac{3}{2}) = 1$ (substitute $y = z^2/2$), using $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$.

(b) $\int e^{tz}\phi(z)\,dz = e^{t^2/2}\int \phi(z-t)\,dz = e^{t^2/2}$ (complete the square). Then $Ee^{t(\sigma Z+\mu)} = e^{\mu t + \sigma^2 t^2/2}$.

(c) Multiply the mgfs. ■

Proposition 3.12 (Representation)

(a) $X \sim N(\mu, \sigma^2) \Rightarrow aX + b \sim N(a\mu + b, a^2\sigma^2)$.
(b) $X \sim N(\mu, \sigma^2) \iff (X-\mu)/\sigma \sim N(0,1) \iff X \stackrel{d}{=} \sigma Z + \mu$ with $Z \sim N(0,1)$.

Proof. $M_{aX+b}(t) = e^{bt} M_X(at) = \exp\{(a\mu + b)t + \frac{1}{2}a^2\sigma^2 t^2\}$;

(b) is the special case $a = 1/\sigma$, $b = -\mu/\sigma$. ■

So $P(X \le x) = \Phi((x-\mu)/\sigma)$, and $\Phi(-z) = 1 - \Phi(z)$ by symmetry. The upper α point z_α ($P(Z > z_\alpha) = \alpha$) gives the upper α point $\mu + \sigma z_\alpha$ of $N(\mu, \sigma^2)$; e.g. $z_{0.05} = 1.645$, $z_{0.025} = 1.96$.

Example 3.5

$X \sim N(10, 9)$. Then $P(7 < X \le 16) = \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = 0.8185$ and $P(X > 13) = 1 - \Phi(1) = 0.1587$. The 95th percentile is $10 + 3(1.645) = 14.94$ and the 2.5th percentile is $10 - 3(1.96) = 4.12$.

The normal density arises as the limit of standardized binomial probabilities (de Moivre–Laplace):

$$\sum_{a \le (x-np)/\sqrt{np(1-p)} \le b} \binom{n}{x} p^x (1-p)^{n-x} \to \int_a^b \phi(z)\, dz.$$

This is a special case of the central limit theorem (Chapter 5).

Summary

distribution	pmf/pdf	representation
$B(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	sum of n i.i.d. Bernoulli(p)
Multi$(n,\mathbf{p})$	$\frac{n!}{\prod x_j!}\prod p_j^{x_j}$	sum of n i.i.d. Multi$(1,\mathbf{p})$
Negbin(r,p)	$\binom{x-1}{r-1}p^r(1-p)^{x-r}$	sum of r i.i.d. Geo(p)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	limit of $B(n,\lambda/n)$
Gamma(r,β)	$\frac{x^{r-1}e^{-x/\beta}}{\Gamma(r)\beta^r}$	sum of r i.i.d. Exp(β)
$N(\mu,\sigma^2)$	$\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$	$\mu+\sigma Z$, $Z\sim N(0,1)$

3.7 Selected exercises

Motivation. These are the course's exercises for Chapter 3 (numbered as there). They give an exponential tail bound for the binomial, characterize the geometric law by memorylessness, justify the binomial approximation of the hypergeometric, and prove the limit $(1 + a_n/n)^n \to e^a$ used in Proposition 3.6.

Exercise 3.5 (A Chernoff bound for the binomial)

Let $X \sim B(n,p)$, $0 < p < 1$, $q = 1 - p$. Show that for $0 < t < nq$,

$$P(X - np \ge t) \le e^{-n h_p(t/n+p)}, \qquad h_p(a) = (1-a)\log\frac{1-a}{1-p} + a\log\frac{a}{p}.$$

Solution.

Step 1. Put $a = p + t/n \in (p, 1)$. For $\lambda > 0$, Theorem 1.10(a) applied to $e^{\lambda X}$ and Proposition 3.2(b) give

$$P(X \ge na) = P(e^{\lambda X} \ge e^{\lambda n a}) \le e^{-\lambda n a}(pe^{\lambda} + q)^n = e^{-n\varphi(\lambda)},$$

where $\varphi(\lambda) = \lambda a - \log(pe^{\lambda} + q)$.

Step 2. $\varphi'(\lambda) = a - pe^{\lambda}/(pe^{\lambda} + q)$ vanishes at $e^{\lambda^*} = aq/\{(1-a)p\}$, and $\lambda^* > 0$ because $aq > (1-a)p$ is equivalent to $a > p$.

Step 3. At λ^*, $pe^{\lambda^*} + q = \frac{aq}{1-a} + q = \frac{q}{1-a}$, so

$$\varphi(\lambda^*) = a\log\frac{aq}{(1-a)p} - \log\frac{q}{1-a} = a\log\frac{a}{p} + (1-a)\log\frac{1-a}{q} = h_p(a).$$

Since $\{X - np \geq t\} = \{X \geq na\}$, the bound follows. ($h_p(a)$ is the Kullback–Leibler divergence of Bernoulli(a) from Bernoulli(p); it is positive for $a \neq p$, so the bound decays exponentially in n.) ■

Exercise 3.9 (Memorylessness characterizes the geometric law)

(a) Let X take values in $\{1, 2, \dots\}$ with $P(X > m) > 0$ and $P(X > m + n \mid X > m) = P(X > n)$ for all positive integers m, n. Show that $X \sim \text{Geo}(p)$ for some $p \in (0, 1)$.
(b) Let $X_i \sim \text{Geo}(p_i)$, $i = 1, \dots, n$, be independent. Show that $\min(X_1, \dots, X_n) \sim \text{Geo}(1 - \prod_i (1 - p_i))$.

Solution.

(a)

Step 1. Let $G(k) = P(X > k)$. The hypothesis says $G(m + n) = G(m)G(n)$ for $m, n \geq 1$. With $q = G(1)$, induction gives $G(k) = q^k$ for $k \geq 1$, and also $G(0) = 1 = q^0$ since $X \geq 1$.

Step 2. $q > 0$ by hypothesis, and $q < 1$: otherwise $G(k) = 1$ for all k, contradicting $G(k) = P(X > k) \to 0$ (Theorem 1.2).

Step 3. Hence $P(X = k) = G(k - 1) - G(k) = q^{k-1}(1 - q)$ for $k \geq 1$, i.e. $X \sim \text{Geo}(p)$ with $p = 1 - q \in (0, 1)$.

(b) With $N = \min_i X_i$ and $q_i = 1 - p_i$, independence gives, for $k \geq 0$,

$$P(N > k) = \prod_{i=1}^{n} P(X_i > k) = \prod_{i=1}^{n} q_i^k = \Big(\prod_{i=1}^{n} q_i\Big)^k.$$

As in Step 3 of (a), this tail determines the law $\text{Geo}(1 - \prod_i q_i)$. (In terms of trials: N is the first time at which at least one of n independent experiments succeeds.) ■

Exercise 3.10 (Hypergeometric to binomial)

Let X_N be hypergeometric with population size N, $D = D_N$ units of type 1 and sample size n (fixed), and let $D_N/N \to p \in [0, 1]$. Show that $P(X_N = x) \to \binom{n}{x} p^x (1 - p)^{n-x}$ for $x = 0, \dots, n$.

Solution.

Step 1. Write $(a)_k = a(a - 1) \cdots (a - k + 1)$. Since $\binom{a}{k} = (a)_k / k!$,

$$P(X_N = x) = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}} = \binom{n}{x} \frac{(D)_x \, (N - D)_{n-x}}{(N)_n}$$

(both sides are 0 when $x > D$ or $n - x > N - D$).

Step 2. Divide numerator and denominator by $N^n = N^x N^{n-x}$. For fixed j, $(D - j)/N \to p$ and $(N - D - j)/N \to 1 - p$, so $(D)_x / N^x \to p^x$ and $(N - D)_{n-x} / N^{n-x} \to (1 - p)^{n-x}$, while $(N)_n / N^n \to 1$. Each is a product of a fixed number of factors, so the limits multiply, and $P(X_N = x) \to \binom{n}{x} p^x (1 - p)^{n-x}$. Sampling without replacement from a large population is nearly sampling with replacement. ■

Exercise 3.15 (A limit of exponential type)

Let $a \in \mathbb{R}$ and $a_n \to a$. Show that $(1 + a_n/n)^n \to e^a$.

Solution.

Step 1. For $|u| \le \frac{1}{2}$, Taylor's theorem gives $\log(1+u) = u + r(u)$ with $|r(u)| \le u^2$: indeed $|r(u)| = |\sum_{k\ge 2} (-1)^{k+1} u^k/k| \le \frac{u^2}{2} \sum_{j\ge 0} |u|^j \le u^2$.

Step 2. (a_n) is bounded, say $|a_n| \le A$, so $|a_n/n| \le \frac{1}{2}$ for $n \ge 2A$. For such n,

$$n \log\Big(1 + \frac{a_n}{n}\Big) = a_n + n\, r\Big(\frac{a_n}{n}\Big), \qquad \Big|n\, r\Big(\frac{a_n}{n}\Big)\Big| \le \frac{a_n^2}{n} \le \frac{A^2}{n} \to 0.$$

Step 3. Hence $n \log(1 + a_n/n) \to a$, and by continuity of the exponential, $(1 + a_n/n)^n \to e^a$. With $a_n = a + o(1)$ this is the claim $\{1 + (a + o(1))/n\}^n \to e^a$. ∎

Chapter 4. Sampling Distributions

4.1 Statistics and sampling distributions

A **random sample** of size n from a population with pdf f is a vector $X_1, \dots, X_n$ of i.i.d. random variables with pdf f. A **statistic** is a function $u(X_1, \dots, X_n)$ of the sample that involves no unknown parameter; its distribution is its **sampling distribution**. Most of statistical inference rests on knowing such distributions, at least approximately.

Example 4.1

For a Bernoulli(p) sample, $n\hat{p} = \sum X_i \sim B(n, p)$, so $E\hat{p} = p$ and $\operatorname{Var}\hat{p} = p(1-p)/n$. The sampling distribution of $\hat{p}$ concentrates around p as n grows, which is what makes $\hat{p}$ a sensible estimate.

Discrete case. If $\mathbf{X}$ is discrete and $Y = u(\mathbf{X})$, then

$$f_Y(y) = P(u(\mathbf{X}) = y) = \sum_{\mathbf{x}:\, u(\mathbf{x})=y} f_{\mathbf{X}}(\mathbf{x}).$$

Example 4.2

Let $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ be independent. Then

$$P(X_1 + X_2 = y) = \sum_{x=0}^{y} \frac{e^{-\lambda_1}\lambda_1^x}{x!}\frac{e^{-\lambda_2}\lambda_2^{y-x}}{(y-x)!} = \frac{e^{-(\lambda_1+\lambda_2)}}{y!}\sum_{x=0}^{y}\binom{y}{x}\lambda_1^x\lambda_2^{y-x} = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^y}{y!},$$

so $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$. The same convolution argument shows $B(n_1, p) \oplus B(n_2, p) = B(n_1 + n_2, p)$, where $\oplus$ denotes the sum of independent variables.

4.2 Transformations of continuous variables

Motivation. Statistics are functions of the data, so we need the distribution of $Y = g(X)$. The cdf method always works; for one-to-one smooth maps the Jacobian formula gives the density directly.

Theorem 4.1 (One-to-one transformations)

Let $\mathbf{X} = (X_1, \dots, X_n)$ have pdf $f_{\mathbf{X}}$ that is positive on an open set $\mathcal{X} \subseteq \mathbb{R}^n$ and zero outside. Let $u : \mathcal{X} \to \mathcal{Y}$ be one-to-one and onto, with inverse $\mathbf{x} = w(\mathbf{y})$ continuously differentiable and Jacobian

$$J(\mathbf{y}) = \det\left(\frac{\partial w_i(\mathbf{y})}{\partial y_j}\right) \neq 0 \quad \text{on } \mathcal{Y}.$$

Then $\mathbf{Y} = u(\mathbf{X})$ has pdf $f_{\mathbf{Y}}(\mathbf{y}) = f_{\mathbf{X}}(w(\mathbf{y}))|J(\mathbf{y})|$ for $\mathbf{y} \in \mathcal{Y}$, and 0 otherwise.

Proof. For a set $B \subseteq \mathcal{Y}$, $\{\mathbf{Y} \in B\} = \{\mathbf{X} \in w(B)\}$, so by the change-of-variables formula for multiple integrals,

$$P(\mathbf{Y} \in B) = \int_{w(B)} f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x} = \int_B f_{\mathbf{X}}(w(\mathbf{y}))|J(\mathbf{y})|\, d\mathbf{y}.$$

Since this holds for every B, the integrand is the pdf of $\mathbf{Y}$. ∎

In one dimension, with u increasing or decreasing, this reads $f_Y(y) = f_X(w(y))|w'(y)|$.

To find the pdf of a single statistic $Y_1 = u_1(\mathbf{X})$, add auxiliary coordinates $Y_2, \dots, Y_n$ so that the full map is one-to-one, apply the theorem, and integrate the auxiliary coordinates out. The choice of auxiliary coordinates is a matter of convenience.

Theorem 4.2 (Many-to-one transformations)

Suppose $\mathcal{X}$ is, up to a set of probability zero, the disjoint union of open sets $\mathcal{X}_1, \dots, \mathcal{X}_m$, and that u restricted to each $\mathcal{X}_j$ is one-to-one from $\mathcal{X}_j$ onto $\mathcal{Y}_j$, with inverse w_j and nonvanishing Jacobian J_j. Then

$$f_{\mathbf{Y}}(\mathbf{y}) = \sum_{j:\, \mathbf{y} \in \mathcal{Y}_j} f_{\mathbf{X}}(w_j(\mathbf{y}))|J_j(\mathbf{y})|.$$

Proof. $P(\mathbf{Y} \in B) = \sum_j P(\mathbf{X} \in \mathcal{X}_j,\ u(\mathbf{X}) \in B) = \sum_j \int_{B \cap \mathcal{Y}_j} f_{\mathbf{X}}(w_j(\mathbf{y}))|J_j(\mathbf{y})|\, d\mathbf{y}$, applying the one-to-one argument of Theorem 4.1 on each piece. ∎

Example 4.3

(a) If $X \sim N(0,1)$ and $Y = X^2$, the two branches are $x = \pm\sqrt{y}$ with $|J| = 1/(2\sqrt{y})$, so

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\Gamma(1/2)2^{1/2}} y^{1/2-1} e^{-y/2}, \qquad y > 0,$$

the Gamma$(1/2, 2)$ density.
(b) If X has pdf $f(x) = (1+x)/2$ on $(-1,1)$ and $Y = X^2$, then for $0 < y < 1$,

$$f_Y(y) = \left\{ \frac{1+\sqrt{y}}{2} + \frac{1-\sqrt{y}}{2} \right\} \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}},$$

which is the Beta$(1/2, 1)$ density. The asymmetry of f is invisible to Y.

Location–scale families. If Z has pdf f, then $X = \mu + \sigma Z$ $(\sigma > 0)$ has pdf $\frac{1}{\sigma} f(\frac{x-\mu}{\sigma})$ by Theorem 4.1. The family $\{\frac{1}{\sigma} f(\frac{x-\mu}{\sigma}) : \mu \in \mathbb{R}, \sigma > 0\}$ is the **location–scale family** generated by f; fixing $\sigma = 1$ gives a location family and fixing $\mu = 0$ a scale family. Standard members:

family	standard pdf $f(z)$	type
normal	$\frac{1}{\sqrt{2\pi}} e^{-z^2/2}$	location–scale
logistic	$e^{-z}/(1+e^{-z})^2$	location–scale
double exponential	$\frac{1}{2} e^{-\lvert z \rvert}$	location–scale
Cauchy	$\frac{1}{\pi} \frac{1}{1+z^2}$	location–scale
exponential	e^{-z}, $z > 0$	scale
gamma (fixed α)	$z^{\alpha-1} e^{-z}/\Gamma(\alpha)$, $z > 0$	scale

In a location–scale family, $(X - \mu)/\sigma$ has a fixed distribution, so its moments and quantiles can be tabulated once. For example, $EX = \mu + \sigma EZ$ and the p-quantile is $\mu + \sigma z_p$. Similarly $U(a,b) \overset{d}{=} a + (b-a)U(0,1)$.

Example 4.4 (Gamma to beta)

Let $X_1 \sim \text{Gamma}(\alpha_1, \beta)$ and $X_2 \sim \text{Gamma}(\alpha_2, \beta)$ be independent, $Y_1 = X_1 + X_2$ and $Y_2 = X_1/(X_1+X_2)$. The map $(0,\infty)^2 \to (0,\infty)\times(0,1)$ has inverse $x_1 = y_1y_2$, $x_2 = y_1(1-y_2)$, with

$$J = \det \begin{pmatrix} y_2 & y_1 \\ 1-y_2 & -y_1 \end{pmatrix} = -y_1.$$

Hence

$$\begin{aligned} f(y_1,y_2) &= \frac{(y_1y_2)^{\alpha_1-1}\{y_1(1-y_2)\}^{\alpha_2-1}e^{-y_1/\beta}}{\Gamma(\alpha_1)\Gamma(\alpha_2)\beta^{\alpha_1+\alpha_2}}\, y_1 \\ &= \frac{y_1^{\alpha_1+\alpha_2-1}e^{-y_1/\beta}}{\Gamma(\alpha_1+\alpha_2)\beta^{\alpha_1+\alpha_2}} \cdot \frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} y_2^{\alpha_1-1}(1-y_2)^{\alpha_2-1}. \end{aligned}$$

The density factorizes, so $Y_1 \sim \text{Gamma}(\alpha_1+\alpha_2, \beta)$ and $Y_2 \sim \text{Beta}(\alpha_1,\alpha_2)$ are independent. Since the second factor must integrate to 1, this also proves

$$B(\alpha_1,\alpha_2) = \int_0^1 y^{\alpha_1-1}(1-y)^{\alpha_2-1}dy = \frac{\Gamma(\alpha_1)\Gamma(\alpha_2)}{\Gamma(\alpha_1+\alpha_2)}.$$

Example 4.5 (Dirichlet distribution)

Let $X_1,\dots,X_{k+1}$ be independent with $X_i \sim \text{Gamma}(\alpha_i,\beta)$, $S = X_1+\cdots+X_{k+1}$, and $Y_i = X_i/S$ for $i = 1,\dots,k$. Use $(Y_1,\dots,Y_k,S)$ as new coordinates: the inverse is $x_i = sy_i$ $(i \le k)$ and $x_{k+1} = s(1-y_1-\cdots-y_k)$. Adding the first k rows of the Jacobian matrix to the last row turns it into $(0,\dots,0,1)$, so $|J| = s^k$. Therefore

$$f(y_1,\dots,y_k,s) = \frac{\prod_{i=1}^k y_i^{\alpha_i-1}\,(1-\sum_{i\le k} y_i)^{\alpha_{k+1}-1}}{\prod_{i=1}^{k+1}\Gamma(\alpha_i)} \cdot \frac{s^{\alpha_1+\cdots+\alpha_{k+1}-1}e^{-s/\beta}}{\beta^{\alpha_1+\cdots+\alpha_{k+1}}}.$$

Integrating out s gives the **Dirichlet**$(\alpha_1,\dots,\alpha_{k+1})$ density

$$f(y_1,\dots,y_k) = \frac{\Gamma(\alpha_1+\cdots+\alpha_{k+1})}{\Gamma(\alpha_1)\cdots\Gamma(\alpha_{k+1})}\prod_{i=1}^k y_i^{\alpha_i-1}\Big(1-\sum_{i=1}^k y_i\Big)^{\alpha_{k+1}-1}$$

on $\{y_i > 0,\ y_1+\cdots+y_k < 1\}$, and shows that $(Y_1,\dots,Y_k)$ is independent of S. For $k=1$ this is $\text{Beta}(\alpha_1,\alpha_2)$.

Example 4.6 (Polar coordinates)

Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} N(0,1)$ and write $Z_1 = R\cos\Theta$, $Z_2 = R\sin\Theta$ with $R > 0$ and $0 < \Theta < 2\pi$. The Jacobian of $(r,\theta) \mapsto (z_1,z_2)$ is r, so

$$f(r,\theta) = \frac{1}{2\pi}e^{-r^2/2}r = (re^{-r^2/2}) \cdot \frac{1}{2\pi}.$$

Thus R and Θ are independent, $\Theta \sim U(0,2\pi)$, and $R^2/2 \sim \text{Exp}(1)$ (equivalently $R^2 \sim \chi^2(2)$). Reading this backwards, if $U_1, U_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, then $\sqrt{-2\log U_1}\cos(2\pi U_2)$ and $\sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent $N(0,1)$ variables.

Example 4.7 (A two-to-one map on the disk)

Let (X_1, X_2) be uniform on the unit disk, with pdf $1/\pi$, and put $Y_1 = X_1^2 + X_2^2$ and $Y_2 = X_1/\sqrt{Y_1}$. The map is two-to-one: on the upper and lower half disks the inverses are

$$x_1 = y_2\sqrt{y_1}, \qquad x_2 = \pm\sqrt{y_1}\sqrt{1-y_2^2}, \qquad 0 < y_1 < 1,\ -1 < y_2 < 1.$$

A direct computation gives $|J| = 1/(2\sqrt{1-y_2^2})$ on each branch. By Theorem 4.2,

$$f(y_1, y_2) = 2 \cdot \frac{1}{\pi} \cdot \frac{1}{2\sqrt{1-y_2^2}} = 1 \cdot \frac{1}{\pi\sqrt{1-y_2^2}}.$$

So $Y_1 \sim U(0,1)$ is independent of Y_2, the cosine of the polar angle, which has the arcsine density $1/(\pi\sqrt{1-y^2})$ on $(-1,1)$.

Example 4.8

For $X_1, X_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, let $Y = (X_1 + X_2)/2$ and $W = (X_1 - X_2)/2$. Then $x_1 = y + w$, $x_2 = y - w$, $|J| = 2$, and (Y, W) is uniform with density 2 on $\{|w| < \min(y, 1-y)\}$. Integrating out w gives the triangular density $f_Y(y) = 4\min(y, 1-y)$ on $(0,1)$.

4.3 Sampling from a normal population

Motivation. For a normal sample, $\bar{X}$ and S^2 are independent, $\bar{X}$ is normal and $(n-1)S^2/\sigma^2$ is χ^2. These facts produce the t and F distributions and with them all exact normal-theory inference.

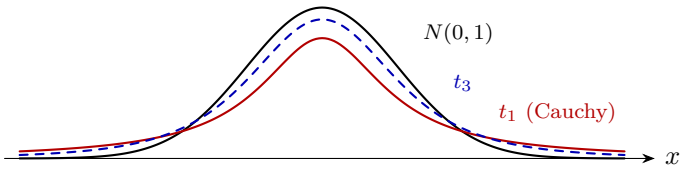


Figure. *The t densities have heavier tails than the normal; replacing σ by S costs exactly this much. As the degrees of freedom grow, $t_\nu \to N(0,1)$.*

4.3.1 The chi-square distribution

Definition (Chi-square)

For $r > 0$, $\chi^2(r) = \text{Gamma}(r/2, 2)$, with pdf

$$f(x) = \frac{1}{\Gamma(r/2)2^{r/2}} x^{r/2-1} e^{-x/2}, \qquad x > 0.$$

r is the number of degrees of freedom. The upper α point $\chi^2_\alpha(r)$ is defined by $P(X > \chi^2_\alpha(r)) = \alpha$.

Theorem 4.3

(a) If $X \sim \chi^2(r)$, then $M(t) = (1-2t)^{-r/2}$ for $t < 1/2$, $EX = r$, and $\operatorname{Var} X = 2r$.

(b) If $X_i \sim \chi^2(r_i)$ are independent, then $X_1 + \cdots + X_k \sim \chi^2(r_1 + \cdots + r_k)$.

(c) If $Z \sim N(0,1)$, then $Z^2 \sim \chi^2(1)$. Hence if $Z_1, \ldots, Z_r \overset{\text{i.i.d.}}{\sim} N(0,1)$, then $Z_1^2 + \cdots + Z_r^2 \sim \chi^2(r)$.

(d) $X \sim \text{Gamma}(\alpha, \beta)$ if and only if $2X/\beta \sim \chi^2(2\alpha)$.

Proof. (a) is the gamma mgf with $\alpha = r/2$, $\beta = 2$, and the moments are $\alpha\beta$ and $\alpha\beta^2$.

(b) The mgf of the sum is $\prod(1-2t)^{-r_i/2} = (1-2t)^{-\sum r_i/2}$.

(c) is Example 4.3(a) combined with (b).

(d) $2X/\beta$ has mgf $(1 - \beta \cdot 2t/\beta)^{-\alpha} = (1-2t)^{-2\alpha/2}$. ■

By (d), gamma quantiles come from chi-square tables: the upper α point of Gamma(ν, β) is $\frac{\beta}{2}\chi^2_\alpha(2\nu)$.

4.3.2 The t distribution

Definition (Student's t)

If $Z \sim N(0,1)$ and $V \sim \chi^2(r)$ are independent, the distribution of $T = Z/\sqrt{V/r}$ is the t distribution with r degrees of freedom, $t(r)$. Its upper α point is $t_\alpha(r)$.

Theorem 4.4

The pdf of $t(r)$ is

$$f(t) = \frac{\Gamma(\frac{r+1}{2})}{\Gamma(\frac{1}{2})\Gamma(\frac{r}{2})\sqrt{r}}\Big(1 + \frac{t^2}{r}\Big)^{-\frac{r+1}{2}}, \qquad t \in \mathbb{R}.$$

It is symmetric about 0; $t(1)$ is the Cauchy distribution. $E|T|^k < \infty$ iff $k < r$, and $ET = 0$ $(r > 1)$, $\operatorname{Var} T = r/(r-2)$ $(r > 2)$. As $r \to \infty$, $f(t) \to \phi(t)$.

Proof. Map $(z, v) \mapsto (t, v)$ with $z = t\sqrt{v/r}$; the Jacobian is $\sqrt{v/r}$. The joint pdf of (T, V) is

$$\frac{1}{\sqrt{2\pi}} e^{-t^2 v/(2r)} \cdot \frac{v^{r/2-1} e^{-v/2}}{\Gamma(r/2) 2^{r/2}} \cdot \sqrt{\frac{v}{r}}.$$

Integrating over $v > 0$ with the gamma integral $\int_0^\infty v^{a-1} e^{-bv} dv = \Gamma(a) b^{-a}$, $a = \frac{r+1}{2}$, $b = \frac{1}{2}(1 + t^2/r)$, gives

$$f(t) = \frac{\Gamma(\frac{r+1}{2}) 2^{(r+1)/2}}{\sqrt{2\pi}\,\Gamma(\frac{r}{2}) 2^{r/2} \sqrt{r}} \Big(1 + \frac{t^2}{r}\Big)^{-\frac{r+1}{2}},$$

which is the stated form since $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. Symmetry is clear, and for $r = 1$ the density is $\frac{1}{\pi}(1+t^2)^{-1}$. The tail behaves like $|t|^{-(r+1)}$, so $|t|^k f(t)$ is integrable iff $k < r$. For the variance, by independence $ET^2 = r\, EZ^2\, EV^{-1}$ and

$$EV^{-1} = \int_0^\infty \frac{v^{r/2-2} e^{-v/2}}{\Gamma(r/2) 2^{r/2}} dv = \frac{\Gamma(r/2 - 1) 2^{r/2-1}}{\Gamma(r/2) 2^{r/2}} = \frac{1}{r-2}.$$

Finally $(1 + t^2/r)^{-(r+1)/2} \to e^{-t^2/2}$, and the constant tends to $1/\sqrt{2\pi}$ (by Stirling's formula, or because both sides are densities and Scheffé-type arguments apply; in Chapter 5 we show $T \to N(0,1)$ in distribution directly). ■

4.3.3 Sample mean and sample variance

Theorem 4.5 (Normal samples)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, $\bar{X} = \frac{1}{n}\sum X_i$ and $S^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2$. Then

(a) $\bar{X} \sim N(\mu, \sigma^2/n)$;

(b) $\bar{X}$ and $(X_1 - \bar{X}, \dots, X_n - \bar{X})$ are independent; in particular $\bar{X}$ and S^2 are independent;

(c) $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$;

(d) $\sqrt{n}(\bar{X} - \mu)/S \sim t(n-1)$.

Proof. (a) and (b). Compute the joint mgf of $\bar{X}$ and the deviations. For $t, t_1, \dots, t_n$ let $\bar{t} = \frac{1}{n}\sum t_i$. Then

$$t\bar{X} + \sum_i t_i(X_i - \bar{X}) = \sum_i a_i X_i, \qquad a_i = \frac{t}{n} + (t_i - \bar{t}),$$

with $\sum a_i = t$ and $\sum a_i^2 = t^2/n + \sum (t_i - \bar{t})^2$ (the cross term vanishes because $\sum (t_i - \bar{t}) = 0$). By independence,

$$E \exp\Big(\sum a_i X_i\Big) = \prod_i \exp\Big(\mu a_i + \frac{\sigma^2 a_i^2}{2}\Big) = \exp\Big(\mu t + \frac{\sigma^2 t^2}{2n}\Big) \cdot \exp\Big(\frac{\sigma^2}{2}\sum_i (t_i - \bar{t})^2\Big).$$

The joint mgf factors into a function of t alone and a function of $(t_1, \dots, t_n)$ alone. Hence $\bar{X}$ is independent of the deviation vector, and setting $t_i = 0$ shows $\bar{X} \sim N(\mu, \sigma^2/n)$. Since S^2 is a function of the deviations, it is independent of $\bar{X}$.

(c) Expand around $\bar{X}$:

$$U := \sum_i \Big(\frac{X_i - \mu}{\sigma}\Big)^2 = \underbrace{\frac{(n-1)S^2}{\sigma^2}}_{V} + \underbrace{\Big(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\Big)^2}_{W}.$$

Here $U \sim \chi^2(n)$ and $W \sim \chi^2(1)$ by Theorem 4.3(c), and V and W are independent by (b). Therefore $M_U(t) = M_V(t) M_W(t)$, i.e.

$$M_V(t) = \frac{(1-2t)^{-n/2}}{(1-2t)^{-1/2}} = (1-2t)^{-(n-1)/2}, \qquad t < \tfrac{1}{2},$$

and $V \sim \chi^2(n-1)$ by uniqueness of mgfs.

(d) $\sqrt{n}(\bar{X} - \mu)/S = Z/\sqrt{V/(n-1)}$ with $Z = \sqrt{n}(\bar{X} - \mu)/\sigma \sim N(0,1)$ independent of $V \sim \chi^2(n-1)$. ■

Corollary 4.6 (Intervals for μ and σ^2)

For a normal sample, each of the following random intervals covers the true parameter with probability $1 - \alpha$:

$$\bar{X} \pm t_{\alpha/2}(n-1)\frac{S}{\sqrt{n}} \quad \text{for } \mu, \qquad \Big(\frac{(n-1)S^2}{\chi^2_{\alpha/2}(n-1)}, \frac{(n-1)S^2}{\chi^2_{1-\alpha/2}(n-1)}\Big) \quad \text{for } \sigma^2.$$

Proof. By Theorem 4.5(d), $P(|\sqrt{n}(\bar{X} - \mu)/S| \le t_{\alpha/2}(n-1)) = 1 - \alpha$, and the event inside is $\{\mu \in \bar{X} \pm t_{\alpha/2} S/\sqrt{n}\}$. Likewise $P(\chi^2_{1-\alpha/2} \le (n-1)S^2/\sigma^2 \le \chi^2_{\alpha/2}) = 1 - \alpha$, and solving for σ^2 gives the second interval. ■

Example 4.9

A sample of $n = 16$ fill weights from a normal population gives $\bar{x} = 52.3$ and $s = 4.8$. With $t_{0.025}(15) = 2.131$, the 95% interval for μ is $52.3 \pm 2.131 \cdot 4.8/4 = 52.3 \pm 2.56$, i.e. $(49.74, 54.86)$. With $\chi^2_{0.025}(15) = 27.49$ and $\chi^2_{0.975}(15) = 6.262$, and $(n-1)s^2 = 345.6$, the interval for σ^2 is $(12.57, 55.19)$.

4.3.4 The F distribution

Definition (Snedecor's F)

If $V_1 \sim \chi^2(r_1)$ and $V_2 \sim \chi^2(r_2)$ are independent, the distribution of $F = (V_1/r_1)/(V_2/r_2)$ is $F(r_1, r_2)$. Its upper α point is $F_\alpha(r_1, r_2)$.

Theorem 4.7

(a) The pdf of $F(r_1, r_2)$ is

$$f(x) = \frac{\Gamma(\frac{r_1+r_2}{2})}{\Gamma(\frac{r_1}{2})\Gamma(\frac{r_2}{2})} \Big(\frac{r_1}{r_2}\Big)^{r_1/2} x^{r_1/2-1} \Big(1 + \frac{r_1}{r_2}x\Big)^{-(r_1+r_2)/2}, \qquad x > 0.$$

(b) If $X \sim F(r_1, r_2)$, then $1/X \sim F(r_2, r_1)$; hence $F_{1-\alpha}(r_1, r_2) = 1/F_\alpha(r_2, r_1)$.

(c) If $T \sim t(r)$, then $T^2 \sim F(1, r)$; hence $t_{\alpha/2}(r)^2 = F_\alpha(1, r)$.

(d) If $X \sim F(r_1, r_2)$, then $\dfrac{(r_1/r_2)X}{1 + (r_1/r_2)X} \sim \text{Beta}(r_1/2, r_2/2)$.

(e) $EX = r_2/(r_2 - 2)$ for $r_2 > 2$.

Proof. (d) first: $\frac{(r_1/r_2)X}{1+(r_1/r_2)X} = \frac{V_1}{V_1+V_2}$, and $V_i \sim \text{Gamma}(r_i/2, 2)$ are independent, so Example 4.4 gives $\text{Beta}(r_1/2, r_2/2)$.

(a) The map $y = cx/(1 + cx)$, $c = r_1/r_2$, is increasing with $dy/dx = c/(1 + cx)^2$, so

$$f(x) = \frac{1}{B(\frac{r_1}{2}, \frac{r_2}{2})} \Big(\frac{cx}{1+cx}\Big)^{r_1/2-1} \Big(\frac{1}{1+cx}\Big)^{r_2/2-1} \frac{c}{(1+cx)^2},$$

which simplifies to the stated pdf.

(b) $1/X = (V_2/r_2)/(V_1/r_1) \sim F(r_2, r_1)$, so $P(X > 1/F_\alpha(r_2, r_1)) = P(1/X < F_\alpha(r_2, r_1)) = 1 - \alpha$.

(c) $T^2 = Z^2/(V/r)$ with $Z^2 \sim \chi^2(1)$ independent of V.

(e) $EX = \frac{r_2}{r_1} EV_1\, EV_2^{-1} = \frac{r_2}{r_1} \cdot r_1 \cdot \frac{1}{r_2-2}$, using EV^{-1} from the proof of Theorem 4.4. ■

Corollary 4.8 (Comparing two variances)

Let $X_{11}, \ldots, X_{1n_1} \overset{\text{i.i.d.}}{\sim} N(\mu_1, \sigma_1^2)$ and $X_{21}, \ldots, X_{2n_2} \overset{\text{i.i.d.}}{\sim} N(\mu_2, \sigma_2^2)$ be independent samples with sample variances S_1^2, S_2^2. Then

$$\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F(n_1 - 1, n_2 - 1),$$

and a $1 - \alpha$ confidence interval for σ_1^2/σ_2^2 is

$$\Big(\frac{S_1^2}{S_2^2} \cdot \frac{1}{F_{\alpha/2}(n_1 - 1, n_2 - 1)},\ \frac{S_1^2}{S_2^2} F_{\alpha/2}(n_2 - 1, n_1 - 1)\Big).$$

Proof. $(n_i-1)S_i^2/\sigma_i^2 \sim \chi^2(n_i-1)$ independently (Theorem 4.5), so the ratio is F by definition. Invert $P(F_{1-\alpha/2} \le \text{ratio} \le F_{\alpha/2}) = 1-\alpha$ and use Theorem 4.7(b) for the lower point. ∎

4.3.5 The one-way model

Consider k independent groups,

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2), \qquad i = 1,\dots,k,\ j = 1,\dots,n_i,$$

with $n = n_1 + \cdots + n_k$. Write $\bar X_i = \frac{1}{n_i}\sum_j X_{ij}$ for the group means, $\bar X = \frac{1}{n}\sum_{i,j} X_{ij} = \sum_i \frac{n_i}{n}\bar X_i$ for the grand mean, and $\bar\mu = \sum_i \frac{n_i}{n}\mu_i$.

Theorem 4.9 (One-way model)

(a) $\bar X_1,\dots,\bar X_k$ are independent, $\bar X_i \sim N(\mu_i, \sigma^2/n_i)$, and they are independent of

$$\text{SSW} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}(X_{ij}-\bar X_i)^2, \qquad \text{SSW}/\sigma^2 \sim \chi^2(n-k).$$

(b) $\dfrac{1}{\sigma^2}\displaystyle\sum_{i=1}^{k} n_i\{\bar X_i - \bar X - (\mu_i-\bar\mu)\}^2 \sim \chi^2(k-1)$, independent of SSW.

(c) Consequently, with $\hat\sigma^2 = \text{SSW}/(n-k)$, if $\mu_1 = \cdots = \mu_k$ then

$$\frac{\sum_i n_i(\bar X_i - \bar X)^2/(k-1)}{\hat\sigma^2} \sim F(k-1, n-k).$$

Proof. (a) Apply Theorem 4.5 to each group: $\bar X_i \sim N(\mu_i,\sigma^2/n_i)$, $\text{SSW}_i/\sigma^2 := \sum_j (X_{ij}-\bar X_i)^2/\sigma^2 \sim \chi^2(n_i-1)$, and $\bar X_i$ is independent of SSW_i. The groups are independent, so all $2k$ quantities $\bar X_1, \text{SSW}_1, \dots, \bar X_k, \text{SSW}_k$ are mutually independent. Adding the chi-squares gives $\text{SSW}/\sigma^2 \sim \chi^2(\sum(n_i-1)) = \chi^2(n-k)$.

(b) Let $Y_i = \sqrt{n_i}(\bar X_i - \mu_i)/\sigma$, so $\mathbf{Y} = (Y_1,\dots,Y_k)' \sim N_k(\mathbf{0}, I)$, and let $\mathbf{w} = (\sqrt{n_1/n},\dots,\sqrt{n_k/n})'$, a unit vector. Since $\bar X - \bar\mu = \sum_i \frac{n_i}{n}(\bar X_i - \mu_i)$, one checks

$$\frac{\sqrt{n_i}}{\sigma}\{\bar X_i - \bar X - (\mu_i - \bar\mu)\} = Y_i - \sqrt{\tfrac{n_i}{n}}\,\mathbf{w}'\mathbf{Y},$$

so the quantity in (b) is $\mathbf{Y}'(I - \mathbf{w}\mathbf{w}')\mathbf{Y}$. The matrix $I - \mathbf{w}\mathbf{w}'$ is symmetric and idempotent with trace $k-1$, so the quadratic form is $\chi^2(k-1)$ by Theorem 4.18 below. It is a function of the group means only, hence independent of SSW by (a).

(c) Under equal means, $\mu_i - \bar\mu = 0$, and the ratio of the two independent chi-squares divided by their degrees of freedom is $F(k-1, n-k)$. ∎

For equal group sizes the argument in (b) can also be done without matrices: if $n_i \equiv m$, then $\bar X_i - \mu_i \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2/m)$ and (b) is Theorem 4.5(c) applied to these k values.

Corollary 4.10 (Two-sample t)

For $k = 2$ with common variance, let $S_p^2 = \text{SSW}/(n_1+n_2-2)$. Then

$$\frac{(\bar X_1 - \bar X_2) - (\mu_1 - \mu_2)}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t(n_1+n_2-2),$$

and $(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2}(n_1+n_2-2)S_p\sqrt{1/n_1+1/n_2}$ is a $1-\alpha$ interval for $\mu_1-\mu_2$.

Proof. $\bar{X}_1 - \bar{X}_2 \sim N(\mu_1-\mu_2, \sigma^2(1/n_1+1/n_2))$ is independent of $\mathrm{SSW}/\sigma^2 \sim \chi^2(n_1+n_2-2)$ by Theorem 4.9(a); apply the definition of t. ■

More generally, for any contrast $\sum c_i\mu_i$, $(\sum c_i\bar{X}_i - \sum c_i\mu_i)/(\hat{\sigma}\sqrt{\sum c_i^2/n_i}) \sim t(n-k)$.

4.4 Order statistics

Let $X_1, \ldots, X_n$ be a random sample from a continuous distribution with pdf f and cdf F. Ties occur with probability 0, so the sorted values $X_{(1)} < X_{(2)} < \cdots < X_{(n)}$, the **order statistics**, are well defined. The sample median, range $X_{(n)} - X_{(1)}$, and extremes are functions of them.

Theorem 4.11

The joint pdf of $(X_{(1)}, \ldots, X_{(n)})$ is

$$f_{(1,\ldots,n)}(y_1, \ldots, y_n) = n!\, f(y_1)f(y_2)\cdots f(y_n), \qquad y_1 < y_2 < \cdots < y_n.$$

Proof. Up to a null set, $\mathbb{R}^n$ is the disjoint union of the $n!$ open sets $\mathcal{X}_\pi = \{x_{\pi(1)} < \cdots < x_{\pi(n)}\}$, π a permutation. On each $\mathcal{X}_\pi$ the sorting map $\mathbf{x} \mapsto (x_{(1)}, \ldots, x_{(n)})$ is the coordinate permutation $\mathbf{x} \mapsto (x_{\pi(1)}, \ldots, x_{\pi(n)})$, one-to-one onto $\{y_1 < \cdots < y_n\}$ with $|J| = 1$. The joint pdf of the sample is symmetric, $\prod f(x_i)$, so each branch contributes $\prod f(y_i)$, and Theorem 4.2 gives $n!\prod f(y_i)$. ■

Theorem 4.12

(a) The cdf of $X_{(r)}$ is $P(X_{(r)} \le y) = \sum_{j=r}^{n} \binom{n}{j} F(y)^j \{1-F(y)\}^{n-j}$, and its pdf is

$$f_{(r)}(y) = \frac{n!}{(r-1)!\,(n-r)!} F(y)^{r-1}\{1-F(y)\}^{n-r} f(y).$$

(b) For $r < s$, the joint pdf of $(X_{(r)}, X_{(s)})$ is, for $y < z$,

$$f_{(r,s)}(y,z) = \frac{n!}{(r-1)!(s-r-1)!(n-s)!} \times F(y)^{r-1}\{F(z)-F(y)\}^{s-r-1}\{1-F(z)\}^{n-s} f(y)f(z).$$

In particular $X_{(n)}$ has cdf F^n, and $X_{(1)}$ has cdf $1-(1-F)^n$.

Proof. (a) $X_{(r)} \le y$ iff at least r observations are $\le y$, and the number $N(y)$ of observations $\le y$ is $B(n, F(y))$. This gives the cdf. Differentiate it, writing $p = F(y)$:

$$\frac{d}{dp}\sum_{j=r}^{n}\binom{n}{j}p^j(1-p)^{n-j} = \sum_{j=r}^{n}\left\{n\tbinom{n-1}{j-1}p^{j-1}(1-p)^{n-j} - n\tbinom{n-1}{j}p^j(1-p)^{n-1-j}\right\}.$$

The sum telescopes: the negative part of the j-th term cancels the positive part of the $(j+1)$-th term, and the last negative term is zero. Only $n\binom{n-1}{r-1}p^{r-1}(1-p)^{n-r}$ remains. Multiply by $f(y)$ (chain rule).

(b) Integrate the joint pdf of Theorem 4.11 over the other coordinates. For $y_r = y$ and $y_s = z$ fixed,

the variables below y contribute

$$\int_{y_1<\cdots<y_{r-1}<y} \prod_{i<r} f(y_i)\,dy_1\cdots dy_{r-1} = \frac{F(y)^{r-1}}{(r-1)!},$$

since by symmetry the unordered integral $F(y)^{r-1}$ is split equally among the $(r-1)!$ orderings. Similarly the variables between y and z give $\{F(z)-F(y)\}^{s-r-1}/(s-r-1)!$, and those above z give $\{1-F(z)\}^{n-s}/(n-s)!$. Multiplying by $n!f(y)f(z)$ gives (b). The same computation with one fixed coordinate reproves (a). ■

The constants have a multinomial reading: for $X_{(r)} \in (y, y+dy]$ we need $r-1$ observations below y, one in $(y, y+dy]$ and $n-r$ above y.

Example 4.10 (Uniform order statistics)

For a $U(0,1)$ sample, $F(y)=y$, so $X_{(r)} \sim \text{Beta}(r, n-r+1)$, with

$$EX_{(r)} = \frac{r}{n+1}, \qquad \text{Var}\, X_{(r)} = \frac{r(n-r+1)}{(n+1)^2(n+2)}.$$

For instance, with $n=5$, $P(X_{(3)} > 0.7)$ is the probability that at most two observations fall below 0.7:

$$\sum_{j=0}^{2} \binom{5}{j} (0.7)^j (0.3)^{5-j} = 0.0024 + 0.0284 + 0.1323 = 0.163.$$

The joint pdf of $(X_{(1)}, X_{(n)})$ is $n(n-1)(z-y)^{n-2}$ on $0<y<z<1$. Changing to $R = X_{(n)} - X_{(1)}$, $M = X_{(1)}$ ($|J|=1$) and integrating m over $(0, 1-r)$ gives the range density $n(n-1)r^{n-2}(1-r)$, i.e. $R \sim \text{Beta}(n-1, 2)$.

Example 4.11 (Uniform spacings)

Let $U_{(1)} < \cdots < U_{(n)}$ be uniform order statistics and $D_1 = U_{(1)}$, $D_i = U_{(i)} - U_{(i-1)}$ $(2 \le i \le n)$. The map is linear and triangular with unit diagonal, so $|J|=1$ and $(D_1, \ldots, D_n)$ has density $n!$ on $\{d_i > 0,\ d_1 + \cdots + d_n < 1\}$. This is the Dirichlet$(1, \ldots, 1)$ density with $n+1$ parameters, since $\Gamma(n+1)/\Gamma(1)^{n+1} = n!$. By Example 4.5, $(D_1, \ldots, D_n) \overset{d}{=} (E_1, \ldots, E_n)/(E_1 + \cdots + E_{n+1})$ for $E_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$. In particular, every spacing (including $1 - U_{(n)}$) has the same Beta$(1, n)$ distribution.

Example 4.12 (Exponential spacings)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(\beta)$, with order statistics $Y_i = X_{(i)}$ and $Y_0 = 0$. Since $\sum_i y_i = \sum_{i=1}^n (n-i+1)(y_i - y_{i-1})$, the joint pdf $n!\beta^{-n} e^{-\sum y_i/\beta}$ becomes, in the variables $W_i = (n-i+1)(Y_i - Y_{i-1})$ (Jacobian of the inverse $1/n!$),

$$f(w_1, \ldots, w_n) = \prod_{i=1}^{n} \frac{1}{\beta} e^{-w_i/\beta}, \qquad w_i > 0.$$

So the normalized spacings $W_1, \ldots, W_n$ are i.i.d. Exp(β). Consequently $X_{(1)} \sim \text{Exp}(\beta/n)$, and

$$X_{(r)} = \sum_{i=1}^{r} \frac{W_i}{n-i+1}, \qquad EX_{(r)} = \beta \sum_{i=1}^{r} \frac{1}{n-i+1}, \qquad \text{Var}\, X_{(r)} = \beta^2 \sum_{i=1}^{r} \frac{1}{(n-i+1)^2}.$$

This is the memoryless property in action: after the $(i-1)$-th failure, the remaining $n-i+1$ lifetimes start afresh.

Theorem 4.13 (Probability integral transform)

Let F be a cdf and $F^{-1}(u) = \inf\{x : F(x) \geq u\}$ for $0 < u < 1$.

(a) If $U \sim U(0,1)$, then $F^{-1}(U) \sim F$.

(b) If $X \sim F$ and F is continuous, then $F(X) \sim U(0,1)$.

Proof. (a) By right continuity, $F(F^{-1}(u)) \geq u$, and $F^{-1}(u) \leq x$ iff $u \leq F(x)$. Hence $P(F^{-1}(U) \leq x) = P(U \leq F(x)) = F(x)$.

(b) For $0 < u < 1$, let $x_u = F^{-1}(u)$. By continuity $F(x_u) = u$, and $F(X) \geq u$ iff $X \geq x_u$ except on a set of probability 0 (where F is flat). So $P(F(X) \geq u) = P(X \geq x_u) = 1 - F(x_u) = 1 - u$, i.e. $P(F(X) < u) = u$ for all $u \in (0,1)$, which identifies $U(0,1)$. ■

Corollary 4.14

Let $U_{(1)} < \cdots < U_{(n)}$ be the order statistics of a $U(0,1)$ sample.

(a) For any cdf F, $(X_{(1)}, \ldots, X_{(n)}) \stackrel{d}{=} (F^{-1}(U_{(1)}), \ldots, F^{-1}(U_{(n)}))$.

(b) If F is continuous, then $(F(X_{(1)}), \ldots, F(X_{(n)})) \stackrel{d}{=} (U_{(1)}, \ldots, U_{(n)})$; in particular $F(X_{(r)}) \sim \text{Beta}(r, n-r+1)$ whatever F is.

Proof. F^{-1} and F are nondecreasing, so they carry the sorted sample to the sorted transformed sample. Apply Theorem 4.13 coordinatewise to the i.i.d. sample and sort. ■

Example 4.13 (Distribution-free interval for a median)

Let m be the median of a continuous F. Then $X_{(r)} < m$ iff at least r observations are below m, and each observation is below m with probability $1/2$. So for $r < s$,

$$P(X_{(r)} < m < X_{(s)}) = \sum_{j=r}^{s-1} \binom{n}{j} 2^{-n},$$

which does not depend on F. With $n = 10$, $r = 2$, $s = 9$, this is $(1024-1-10-10-1)/1024 = 0.979$.

4.5 The multivariate normal distribution

Throughout, vectors are columns, $\mathbf{Z} = (Z_1, \ldots, Z_m)'$ has i.i.d. $N(0,1)$ entries, written $\mathbf{Z} \sim N_m(\mathbf{0}, I)$, and Σ denotes a symmetric nonnegative definite matrix.

Definition (Multivariate normal)

A random vector $\mathbf{X} \in \mathbb{R}^k$ has the **multivariate normal** distribution $N_k(\mu, \Sigma)$ if any of the following equivalent conditions holds:

(1) $\mathbf{X} \stackrel{d}{=} A\mathbf{Z} + \mu$ for some $k \times m$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_m(\mathbf{0}, I)$;

(2) the mgf of $\mathbf{X}$ is $M(\mathbf{t}) = \exp\left(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}\right)$ for all $\mathbf{t} \in \mathbb{R}^k$;

(3) for every $\mathbf{t}$, $\mathbf{t}'\mathbf{X} \sim N(\mathbf{t}'\mu, \mathbf{t}'\Sigma\mathbf{t})$;

(4) (when Σ is nonsingular) $\mathbf{X}$ has pdf

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{k/2}(\det\Sigma)^{1/2}} \exp\Big\{-\frac{1}{2}(\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)\Big\}.$$

Theorem 4.15

The four conditions are equivalent (the fourth under $\det\Sigma > 0$). Moreover, if $\mathbf{X} \sim N_k(\mu, \Sigma)$, then $E\mathbf{X} = \mu$ and $\operatorname{Var}\mathbf{X} = \Sigma$.

Proof. (1)⇒(2): $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}Ee^{(A'\mathbf{t})'\mathbf{Z}} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, using the mgf $e^{\mathbf{s}'\mathbf{s}/2}$ of $\mathbf{Z}$.

(2)⇒(3): the mgf of $\mathbf{t}'\mathbf{X}$ at s is $M(s\mathbf{t}) = \exp(s\mathbf{t}'\mu + \frac{1}{2}s^2\mathbf{t}'\Sigma\mathbf{t})$, the $N(\mathbf{t}'\mu, \mathbf{t}'\Sigma\mathbf{t})$ mgf.

(3)⇒(2): $M(\mathbf{t}) = Ee^{\mathbf{t}'\mathbf{X}}$ is the mgf of the normal variable $\mathbf{t}'\mathbf{X}$ evaluated at 1, i.e. $\exp(\mathbf{t}'\mu + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$; here μ and Σ are the mean and covariance of $\mathbf{X}$, because $E\mathbf{t}'\mathbf{X} = \mathbf{t}'E\mathbf{X}$ and $\operatorname{Var}\mathbf{t}'\mathbf{X} = \mathbf{t}'\operatorname{Var}(\mathbf{X})\mathbf{t}$ for all $\mathbf{t}$.

(2)⇒(1): take $A = \Sigma^{1/2}$ (or any A with $AA' = \Sigma$); then $A\mathbf{Z} + \mu$ has the same mgf, and mgfs determine distributions.

(1)⇔(4): if Σ is nonsingular, take $A = \Sigma^{1/2}$, which is invertible. The map $\mathbf{z} \mapsto \mathbf{x} = A\mathbf{z} + \mu$ has inverse Jacobian $\det A^{-1} = (\det\Sigma)^{-1/2}$, and $\mathbf{z}'\mathbf{z} = (\mathbf{x}-\mu)'\Sigma^{-1}(\mathbf{x}-\mu)$. Theorem 4.1 applied to $\prod\phi(z_i) = (2\pi)^{-k/2}e^{-\mathbf{z}'\mathbf{z}/2}$ gives the pdf in (4). Conversely a vector with pdf (4) has the same distribution as $\Sigma^{1/2}\mathbf{Z} + \mu$.

Finally, the cgf $\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t}$ has gradient μ and Hessian Σ at $\mathbf{t} = \mathbf{0}$. ■

Example 4.14 (Bivariate normal)

For $k = 2$, write $\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$ with $|\rho| < 1$. Then $\det\Sigma = \sigma_1^2\sigma_2^2(1-\rho^2)$, and with $z_i = (x_i - \mu_i)/\sigma_i$,

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\Big\{-\frac{z_1^2 - 2\rho z_1 z_2 + z_2^2}{2(1-\rho^2)}\Big\}.$$

Completing the square, $z_1^2 - 2\rho z_1 z_2 + z_2^2 = (1-\rho^2)z_1^2 + (z_2 - \rho z_1)^2$, so f is the $N(\mu_1, \sigma_1^2)$ density of x_1 times the $N\big(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x_1 - \mu_1),\ \sigma_2^2(1-\rho^2)\big)$ density of x_2.

Theorem 4.16 (Properties)

Let $\mathbf{X} \sim N_k(\mu, \Sigma)$.

(a) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_k(\mathbf{0}, I)$.

(b) For a $p \times k$ matrix B and $\mathbf{b} \in \mathbb{R}^p$, $B\mathbf{X} + \mathbf{b} \sim N_p(B\mu + \mathbf{b}, B\Sigma B')$.

(c) If $\mathbf{X}_1 \sim N_k(\mu_1, \Sigma_1)$ and $\mathbf{X}_2 \sim N_k(\mu_2, \Sigma_2)$ are independent, then $\mathbf{X}_1 + \mathbf{X}_2 \sim N_k(\mu_1 + \mu_2, \Sigma_1 + \Sigma_2)$.

(d) For matrices A $(p \times k)$ and B $(q \times k)$, $A\mathbf{X}$ and $B\mathbf{X}$ are independent iff $\operatorname{Cov}(A\mathbf{X}, B\mathbf{X}) = A\Sigma B' = 0$.

Proof. (a) is (2)⇒(1) above.

(b) $B\mathbf{X} + \mathbf{b} \overset{d}{=} (BA)\mathbf{Z} + (B\mu + \mathbf{b})$ with $(BA)(BA)' = B\Sigma B'$.

(c) Multiply the mgfs.

(d) Independence always implies zero covariance. Conversely, the stacked vector $\binom{A\mathbf{X}}{B\mathbf{X}}$ is normal by (b) with covariance $\begin{pmatrix} A\Sigma A' & A\Sigma B' \\ B\Sigma A' & B\Sigma B' \end{pmatrix}$. If $A\Sigma B' = 0$, then its mgf at $(\mathbf{s}, \mathbf{t})$ is

$$\exp\left(\mathbf{s}'A\mu + \tfrac{1}{2}\mathbf{s}'A\Sigma A'\mathbf{s}\right) \cdot \exp\left(\mathbf{t}'B\mu + \tfrac{1}{2}\mathbf{t}'B\Sigma B'\mathbf{t}\right),$$

the product of the two marginal mgfs. So $A\mathbf{X}$ and $B\mathbf{X}$ are independent. ■

Taking A and B to be coordinate selections: subvectors of a multivariate normal vector are independent iff their cross-covariance block is zero. This needs joint normality. If $Z \sim N(0,1)$ and $W = SZ$ with an independent random sign $S = \pm 1$, then $W \sim N(0,1)$ and $\operatorname{Cov}(Z, W) = 0$, but $|W| = |Z|$, so Z and W are dependent; (Z, W) is not bivariate normal, since $P(Z + W = 0) = 1/2$.

Theorem 4.17 (Marginal and conditional distributions)

Partition

$$\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right),$$

with Σ_{22} nonsingular. Then

(a) $\mathbf{X}_1 \sim N(\mu_1, \Sigma_{11})$ and $\mathbf{X}_2 \sim N(\mu_2, \Sigma_{22})$;

(b) $\mathbf{X}_1 \mid \mathbf{X}_2 = \mathbf{x}_2 \sim N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{x}_2 - \mu_2),\ \Sigma_{11\cdot 2})$, where $\Sigma_{11\cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$.

Proof. (a) Apply Theorem 4.16(b) with $B = (I\ 0)$ or $(0\ I)$.

(b) Use the block transformation

$$\begin{pmatrix} \mathbf{W} \\ \mathbf{X}_2 \end{pmatrix} = \begin{pmatrix} I & -\Sigma_{12}\Sigma_{22}^{-1} \\ 0 & I \end{pmatrix} \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}.$$

By Theorem 4.16(b), $(\mathbf{W}, \mathbf{X}_2)$ is normal with

$$\operatorname{Cov}(\mathbf{W}, \mathbf{X}_2) = \Sigma_{12} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22} = 0, \qquad \operatorname{Var}\mathbf{W} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

So $\mathbf{W} \sim N(\mu_1 - \Sigma_{12}\Sigma_{22}^{-1}\mu_2, \Sigma_{11\cdot 2})$ is independent of $\mathbf{X}_2$ by part (d) of that theorem. Since $\mathbf{X}_1 = \mathbf{W} + \Sigma_{12}\Sigma_{22}^{-1}\mathbf{X}_2$, conditioning on $\mathbf{X}_2 = \mathbf{x}_2$ only shifts the (unchanged) law of $\mathbf{W}$ by $\Sigma_{12}\Sigma_{22}^{-1}\mathbf{x}_2$. ■

Two features stand out: the conditional mean is linear in $\mathbf{x}_2$ (the regression of $\mathbf{X}_1$ on $\mathbf{X}_2$ is linear), and the conditional covariance does not depend on $\mathbf{x}_2$.

Example 4.15

Let $(X_1, X_2, X_3)' \sim N_3(\mathbf{0}, \Sigma)$ with

$$\Sigma = \begin{pmatrix} 4 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

For the law of X_1 given $(X_2, X_3) = (x_2, x_3)$: $\Sigma_{12} = (2, 1)$ and $\Sigma_{22}^{-1} = \frac{1}{5}\begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$, so $\Sigma_{12}\Sigma_{22}^{-1} = \frac{1}{5}(3, 1)$. The conditional mean is $(3x_2 + x_3)/5$ and the conditional variance is $4 - \frac{1}{5}(3 \cdot 2 + 1 \cdot 1) = 13/5$. Also $X_1 - X_2$ and $X_1 + X_2 - 3X_3$ are independent iff $(1, -1, 0)\Sigma(1, 1, -3)' = 0$; here $\Sigma(1, 1, -3)' = (3, 2, -4)'$, and the inner product is $3 - 2 = 1 \neq 0$, so they are dependent.

4.5.1 Quadratic forms

Theorem 4.18

(a) If $\mathbf{X} \sim N_k(\mu, \Sigma)$ with Σ nonsingular, then $(\mathbf{X}-\mu)'\Sigma^{-1}(\mathbf{X}-\mu) \sim \chi^2(k)$.

(b) Let $\mathbf{Z} \sim N_n(\mathbf{0}, I)$ and A symmetric. Then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ if and only if $A^2 = A$, and in that case $r = \operatorname{rank} A = \operatorname{tr} A$.

(c) If A and B are symmetric idempotent with $AB = 0$, then $\mathbf{Z}'A\mathbf{Z}$ and $\mathbf{Z}'B\mathbf{Z}$ are independent.

Proof. (a) $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X}-\mu) \sim N_k(\mathbf{0}, I)$ and the form equals $\mathbf{Z}'\mathbf{Z} = \sum Z_i^2$.

(b) Diagonalize $A = P'\Lambda P$ with P orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Then $\mathbf{Y} = P\mathbf{Z} \sim N_n(\mathbf{0}, PP') = N_n(\mathbf{0}, I)$ and $\mathbf{Z}'A\mathbf{Z} = \sum \lambda_i Y_i^2$, whose mgf is $\prod_i (1-2\lambda_i t)^{-1/2}$ near $t = 0$. If $A^2 = A$, every $\lambda_i \in \{0, 1\}$, the number of ones is $\operatorname{tr} A = \operatorname{rank} A = r$, and the form is a sum of r squared independent standard normals. Conversely, if the form is $\chi^2(r)$, then $\prod_i (1-2\lambda_i t) = (1-2t)^r$ for all t near 0. Both sides are polynomials in t, so they agree everywhere and have the same roots: every nonzero λ_i equals 1, and there are r of them. Then $\Lambda^2 = \Lambda$ and $A^2 = A$.

(c) $A\mathbf{Z}$ and $B\mathbf{Z}$ are jointly normal with $\operatorname{Cov}(A\mathbf{Z}, B\mathbf{Z}) = AB' = AB = 0$, hence independent by Theorem 4.16(d). Since A is symmetric idempotent, $\mathbf{Z}'A\mathbf{Z} = (A\mathbf{Z})'(A\mathbf{Z})$ is a function of $A\mathbf{Z}$; likewise for B. ∎

The ellipsoid $\{\mu : (\mathbf{X}-\mu)'\Sigma^{-1}(\mathbf{X}-\mu) \le \chi^2_\alpha(k)\}$ is therefore a $1-\alpha$ confidence region for μ when Σ is known.

Example 4.16 (Sample variance revisited)

Let $\mathbf{1}$ be the vector of ones and $J_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, the projection onto the span of $\mathbf{1}$. For $\mathbf{X} \sim N_n(\mu\mathbf{1}, \sigma^2 I)$, put $\mathbf{Z} = (\mathbf{X}-\mu\mathbf{1})/\sigma$. Then

$$\frac{(n-1)S^2}{\sigma^2} = \mathbf{Z}'(I - J_n)\mathbf{Z}, \qquad \frac{n(\bar{X}-\mu)^2}{\sigma^2} = \mathbf{Z}'J_n\mathbf{Z},$$

because $(I-J_n)\mathbf{1} = \mathbf{0}$. Both matrices are symmetric idempotent, $(I-J_n)J_n = 0$, and $\operatorname{tr}(I-J_n) = n-1$. Theorem 4.18 gives parts (b) and (c) of Theorem 4.5 again.

4.5.2 The normal linear model

Let $\mathbf{Y} = X\beta + \mathbf{e}$, where X is a known $n \times p$ matrix of rank p, $\beta \in \mathbb{R}^p$ is unknown, and $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$. The least squares estimator and the residual variance estimator are

$$\hat{\beta} = (X'X)^{-1}X'\mathbf{Y}, \qquad \hat{\sigma}^2 = \frac{1}{n-p}\|\mathbf{Y} - X\hat{\beta}\|^2.$$

Theorem 4.19

(a) $\hat{\beta} \sim N_p(\beta, \sigma^2(X'X)^{-1})$.

(b) $(n-p)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-p)$, independent of $\hat{\beta}$.

(c) For each j, with c_{jj} the j-th diagonal entry of $(X'X)^{-1}$, $(\hat{\beta}_j - \beta_j)/(\hat{\sigma}\sqrt{c_{jj}}) \sim t(n-p)$.

Proof. Let $H = X(X'X)^{-1}X'$, the projection onto the column space of X; it is symmetric, idempotent, of trace $\operatorname{tr}\{(X'X)^{-1}X'X\} = p$, and $HX = X$.

(a) $\hat\beta = (X'X)^{-1}X'\mathbf{Y}$ is linear in $\mathbf{Y} \sim N(X\beta, \sigma^2 I)$, with mean $(X'X)^{-1}X'X\beta = \beta$ and covariance $\sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) The residual vector is $\mathbf{Y} - X\hat\beta = (I-H)\mathbf{Y} = (I-H)\mathbf{e}$, since $(I-H)X = 0$. Hence $(n-p)\hat\sigma^2/\sigma^2 = \mathbf{Z}'(I-H)\mathbf{Z}$ with $\mathbf{Z} = \mathbf{e}/\sigma \sim N_n(\mathbf{0}, I)$, and $I - H$ is idempotent with trace $n - p$; Theorem 4.18(b) applies. For independence, $\operatorname{Cov}\left((X'X)^{-1}X'\mathbf{Y}, (I-H)\mathbf{Y}\right) = \sigma^2(X'X)^{-1}X'(I-H) = 0$, so $\hat\beta$ is independent of the residual vector (Theorem 4.16(d)), hence of $\hat\sigma^2$.

(c) Combine (a) and (b) in the definition of t. ■

Example 4.17 (Simple linear regression)

For $Y_i = \beta_0 + \beta_1 x_i + e_i$, the estimators are

$$\hat\beta_1 = \frac{S_{xY}}{S_{xx}}, \qquad \hat\beta_0 = \bar Y - \hat\beta_1 \bar x, \qquad \hat\beta_1 \sim N\Big(\beta_1, \frac{\sigma^2}{S_{xx}}\Big),$$

where $S_{xx} = \sum (x_i - \bar x)^2$ and $S_{xY} = \sum (x_i - \bar x)(Y_i - \bar Y)$. The residual estimator is $\hat\sigma^2 = (S_{YY} - S_{xY}^2/S_{xx})/(n-2)$, with $S_{YY} = \sum (Y_i - \bar Y)^2$. By Theorem 4.19,

$$\frac{\hat\beta_1 - \beta_1}{\hat\sigma / \sqrt{S_{xx}}} \sim t(n-2).$$

With $n = 10$, $S_{xx} = 40$, $\hat\beta_1 = 1.5$ and $\hat\sigma^2 = 2.0$, the standard error is $\sqrt{2/40} = 0.224$, and with $t_{0.025}(8) = 2.306$ the 95% interval for the slope is 1.5 ± 0.52.

Summary of sampling distributions

distribution	representation
$\chi^2(r) = \text{Gamma}(r/2, 2)$	$Z_1^2 + \cdots + Z_r^2$, $Z_i \overset{\text{i.i.d.}}{\sim} N(0,1)$
$t(r)$	$Z/\sqrt{V/r}$, $Z \sim N(0,1) \perp V \sim \chi^2(r)$
$F(r_1, r_2)$	$(V_1/r_1)/(V_2/r_2)$, $V_i \sim \chi^2(r_i)$ independent
$\text{Beta}(\alpha_1, \alpha_2)$	$Z_1/(Z_1 + Z_2)$, $Z_i \sim \text{Gamma}(\alpha_i, \beta)$ independent
$\text{Dirichlet}(\alpha_1, \ldots, \alpha_{k+1})$	$(Z_1, \ldots, Z_k)/(Z_1 + \cdots + Z_{k+1})$, $Z_i \sim \text{Gamma}(\alpha_i, \beta)$
$N_k(\mu, \Sigma)$	$\Sigma^{1/2}\mathbf{Z} + \mu$, $\mathbf{Z} \sim N_k(\mathbf{0}, I)$

Chapter 5. Approximating Sampling Distributions

Aim

Exact sampling distributions are available only in special cases such as normal samples (Chapter 4). For other cases we approximate them by their limits as $n \to \infty$. The tools are the central limit theorem, the law of large numbers, Slutsky's theorem and the delta method. A last section approximates distributions by simulation.

5.1 The central limit theorem

Already in Chapter 3 we saw the normal approximation to the binomial. If $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $\hat{p} = \bar{X}_n$, then

$$\lim_{n\to\infty} P\Big(a \le \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \le b\Big) = P(a \le Z \le b), \qquad Z \sim N(0,1).$$

The central limit theorem says that the same holds for the sample mean of any population with finite variance.

Theorem 5.1 (Central limit theorem)

Let $X_1, \dots, X_n$ be i.i.d. with $EX_1 = \mu$ and $\operatorname{Var} X_1 = \sigma^2 \in (0, \infty)$. Then for every x,

$$\lim_{n\to\infty} P\Big(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \le x\Big) = \Phi(x).$$

Proof. The theorem holds as stated, but its general proof is beyond our level. We assume in addition that X_1 has an mgf near 0, and use the following fact (the *continuity theorem*): if $M_{Z_n}(t) \to M_Z(t)$ for all $|t| < h$ and Z has a continuous cdf, then $P(Z_n \le x) \to P(Z \le x)$ for all x.

Let $m(s)$ be the mgf of $(X_1 - \mu)/\sigma$, so $m(0) = 1$, $m'(0) = 0$, $m''(0) = 1$. Since the summands are i.i.d.,

$$M_{\sqrt{n}(\bar{X}_n - \mu)/\sigma}(t) = E \exp\Big\{\frac{t}{\sqrt{n}} \sum_{i=1}^{n} \frac{X_i - \mu}{\sigma}\Big\} = \Big\{m\Big(\frac{t}{\sqrt{n}}\Big)\Big\}^n.$$

By Taylor's theorem, $m(s) = 1 + \frac{s^2}{2} + R(s)$ with $R(s)/s^2 \to 0$ as $s \to 0$. Hence

$$m\Big(\frac{t}{\sqrt{n}}\Big) = 1 + \frac{t^2}{2n} + R_n, \qquad nR_n = t^2 \frac{R(t/\sqrt{n})}{(t/\sqrt{n})^2} \to 0.$$

Using $\log(1 + x) = x + o(x)$ as $x \to 0$,

$$\log M_{\sqrt{n}(\bar{X}_n - \mu)/\sigma}(t) = n \log\Big(1 + \frac{t^2}{2n} + R_n\Big) = n\Big(\frac{t^2}{2n} + r_n\Big), \qquad nr_n \to 0,$$

so the mgf tends to $e^{t^2/2}$, the mgf of $N(0,1)$. The continuity theorem finishes the proof. ∎

In practice we read the theorem as $P(a < \bar{X}_n \le b) \approx \Phi(\frac{b-\mu}{\sigma/\sqrt{n}}) - \Phi(\frac{a-\mu}{\sigma/\sqrt{n}})$ for large n. The convergence of the cdfs is in fact uniform in x, which justifies replacing fixed a, b by values that depend on n.

Example 5.1 (Poisson and gamma)

(a) If $Y \sim \text{Poisson}(\lambda n)$, then $Y \overset{d}{=} X_1 + \cdots + X_n$ with $X_i \overset{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, so $(Y - n\lambda)/\sqrt{n\lambda}$ is approximately $N(0,1)$. Thus a Poisson variable with a large mean ν is approximately $N(\nu, \nu)$. For $Y \sim \text{Poisson}(100)$, $P(90 < Y \le 110) \approx \Phi(1) - \Phi(-1) = 0.683$.
(b) Likewise $\text{Gamma}(n\alpha, \beta)$ is the law of a sum of n i.i.d. $\text{Gamma}(\alpha, \beta)$ variables, so for large shape it is approximately normal with the same mean and variance. In particular $\chi^2(r) \approx N(r, 2r)$ for large r. For $Y \sim \chi^2(50)$, $P(Y > 60) \approx 1 - \Phi(10/10) = 0.159$, against the exact value 0.155.

The theorem extends to random vectors; the proof is the same, with the joint mgf in place of the mgf.

Theorem 5.2 (Multivariate central limit theorem)

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be i.i.d. random vectors in $\mathbb{R}^k$ with mean μ and covariance matrix Σ. Then for every $\mathbf{x} \in \mathbb{R}^k$,

$$\lim_{n\to\infty} P(\sqrt{n}(\bar{\mathbf{X}}_n - \mu) \le \mathbf{x}) = P(\mathbf{Z} \le \mathbf{x}), \qquad \mathbf{Z} \sim N_k(\mathbf{0}, \Sigma),$$

where the inequalities are componentwise. (If Σ is singular, the statement holds at all $\mathbf{x}$ where the limit cdf is continuous.)

Example 5.2 (Multinomial)

Let $\mathbf{X}_n = (X_{n1}, \ldots, X_{nk})' \sim \text{Multi}(n, \mathbf{p})$ with $\mathbf{p} = (p_1, \ldots, p_k)'$. By the representation of Chapter 3, $\mathbf{X}_n \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with $\mathbf{Z}_i \overset{\text{i.i.d.}}{\sim} \text{Multi}(1, \mathbf{p})$, whose mean is $\mathbf{p}$ and covariance is $D(\mathbf{p}) - \mathbf{p}\mathbf{p}'$, where $D(\mathbf{p})$ is the diagonal matrix with entries p_j. Hence

$$\frac{\mathbf{X}_n - n\mathbf{p}}{\sqrt{n}} \text{ is approximately } N_k(\mathbf{0},\ D(\mathbf{p}) - \mathbf{p}\mathbf{p}').$$

This covariance matrix is singular: $(D(\mathbf{p}) - \mathbf{p}\mathbf{p}')\mathbf{1} = \mathbf{p} - \mathbf{p} = \mathbf{0}$, which reflects $\sum_j X_{nj} = n$.

5.2 Limiting distributions and convergence in probability

Motivation. Exact sampling distributions are rarely available. As $n \to \infty$ many statistics become approximately normal or concentrate at a constant, and these two modes of convergence are the language of large-sample statistics.

Example 5.3

Let $X_n \equiv 1/n$. Its cdf $F_n(x) = \mathbf{1}\{x \ge 1/n\}$ converges to $G(x) = \mathbf{1}\{x > 0\}$. G is not right-continuous at 0, so it is not a cdf. Still, it agrees with the cdf $F(x) = \mathbf{1}\{x \ge 0\}$ of the constant 0 except at $x = 0$, the discontinuity point of F. It is natural to say that X_n is approximately distributed as the constant 0, and to ignore discontinuity points of the limit.

Definition (Limiting distribution)

Let X_n $(n = 1, 2, \dots)$ and Z be random variables, and let $C(F_Z)$ be the set of continuity points of F_Z. If

$$\lim_{n\to\infty} P(X_n \le x) = P(Z \le x) \qquad \text{for all } x \in C(F_Z),$$

then the distribution of Z is the **limiting** (or asymptotic) distribution of X_n, and X_n **converges in distribution** to Z, written $X_n \xrightarrow{d} Z$.

The CLT says $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} N(0,1)$.

Example 5.4

(a) For $X_n \sim B(n, \lambda/n)$ and $X \sim \text{Poisson}(\lambda)$, Chapter 3 showed $P(X_n = k) \to P(X = k)$ for every k. Both are supported on $\{0, 1, 2, \dots\}$, so $P(X_n \le x) = \sum_{0\le k\le x} P(X_n = k) \to P(X \le x)$ for all x (a finite sum). Hence $X_n \xrightarrow{d} \text{Poisson}(\lambda)$.
(b) Let $U_{(1)}$ be the minimum of n i.i.d. $U(0,1)$ variables. For $0 \le x \le n$,

$$P(nU_{(1)} > x) = \left(1 - \frac{x}{n}\right)^n \to e^{-x},$$

so $nU_{(1)} \xrightarrow{d} \text{Exp}(1)$. The same computation gives $n(1 - U_{(n)}) \xrightarrow{d} \text{Exp}(1)$.

Theorem 5.3 (Convergence to a constant)

Let c be a constant. Then $X_n \xrightarrow{d} c$ (i.e. to a variable X with $P(X = c) = 1$) if and only if

$$\lim_{n\to\infty} P(|X_n - c| \ge \epsilon) = 0 \qquad \text{for every } \epsilon > 0.$$

Proof. The cdf of the constant c is $\mathbf{1}\{x \ge c\}$, continuous everywhere except at c.

($\Rightarrow$) Fix $\epsilon > 0$. Since $c - \epsilon$ and $c + \epsilon/2$ are continuity points,

$$P(X_n \le c - \epsilon) \to 0, \qquad P(X_n \ge c + \epsilon) \le P(X_n > c + \epsilon/2) = 1 - P(X_n \le c + \epsilon/2) \to 0.$$

Adding, $P(|X_n - c| \ge \epsilon) \to 0$.

($\Leftarrow$) For $x < c$, put $\epsilon = c - x$: $0 \le P(X_n \le x) \le P(|X_n - c| \ge \epsilon) \to 0$. For $x > c$, put $\epsilon = x - c$: $0 \le 1 - P(X_n \le x) = P(X_n > x) \le P(|X_n - c| \ge \epsilon) \to 0$. So $P(X_n \le x) \to \mathbf{1}\{x \ge c\}$ for all $x \ne c$. ■

Definition (Convergence in probability)

X_n **converges in probability** to the constant c, written $X_n \xrightarrow{p} c$ or $\operatorname{plim} X_n = c$, if $\lim_n P(|X_n - c| \ge \epsilon) = 0$ for every $\epsilon > 0$. For random vectors, $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ means $P(\|\mathbf{X}_n - \mathbf{c}\| \ge \epsilon) \to 0$ for every $\epsilon > 0$, with $\|\cdot\|$ the Euclidean norm.

By Theorem 5.3, convergence in probability to a constant is the same as convergence in distribution to that constant. An estimator that converges in probability to its target is called **consistent**.

Theorem 5.4 (Weak law of large numbers)

If $X_1, \dots, X_n$ are i.i.d. and $EX_1 = \mu$ is finite, then $\bar{X}_n \xrightarrow{p} \mu$.

Proof. The general proof is beyond our level; we assume in addition $\sigma^2 = \operatorname{Var} X_1 < \infty$. By Chebyshev's inequality (Theorem 1.10), for every $\epsilon > 0$,

$$0 \le P(|\bar{X}_n - \mu| \ge \epsilon) \le \frac{\operatorname{Var} \bar{X}_n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2} \to 0.$$

■

Applied to indicators $\mathbf{1}_A(X_i)$, the law says that the relative frequency of an event converges in probability to its probability:

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_A(X_i) \xrightarrow{p} E\mathbf{1}_A(X_1) = P(X_1 \in A).$$

This is the frequency interpretation of probability, and the reason a sample mean is a sensible estimate of a population mean.

Theorem 5.5

For random vectors $\mathbf{X}_n = (X_{n1}, \ldots, X_{nk})'$ and $\mathbf{c} = (c_1, \ldots, c_k)'$, the following are equivalent:

(a) $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$;

(b) $P(\max_{1 \le i \le k} |X_{ni} - c_i| \ge \epsilon) \to 0$ for every $\epsilon > 0$;

(c) $X_{ni} \xrightarrow{p} c_i$ for each $i = 1, \ldots, k$.

Proof. (a)⇔(b): From $\max_i |x_i| \le \|\mathbf{x}\| \le \sqrt{k} \max_i |x_i|$,

$$P(\max_i |X_{ni} - c_i| \ge \epsilon) \le P(\|\mathbf{X}_n - \mathbf{c}\| \ge \epsilon) \le P(\max_i |X_{ni} - c_i| \ge \epsilon/\sqrt{k}).$$

(b)⇔(c): Since $\{|X_{ni} - c_i| \ge \epsilon\} \subseteq \{\max_j |X_{nj} - c_j| \ge \epsilon\} = \bigcup_j \{|X_{nj} - c_j| \ge \epsilon\}$,

$$\max_i P(|X_{ni} - c_i| \ge \epsilon) \le P(\max_i |X_{ni} - c_i| \ge \epsilon) \le \sum_{i=1}^{k} P(|X_{ni} - c_i| \ge \epsilon).$$

■

Corollary 5.6 (Multivariate law of large numbers)

If $\mathbf{X}_1, \ldots, \mathbf{X}_n$ are i.i.d. random vectors with finite mean μ, then $\bar{\mathbf{X}}_n \xrightarrow{p} \mu$.

Proof. Apply Theorem 5.4 to each coordinate and use Theorem 5.5. ■

Example 5.5 (Sample moments)

The r-th **sample moment** $\hat{m}_r = \frac{1}{n}\sum X_i^r$ estimates the population moment $m_r = EX_1^r$. If $E|X_1|^k < \infty$, all lower moments are finite too, and Corollary 5.6 applied to $(X_i, X_i^2, \ldots, X_i^k)$ gives

$$(\hat{m}_1, \ldots, \hat{m}_k)' \xrightarrow{p} (m_1, \ldots, m_k)'.$$

Theorem 5.7 (Continuous functions preserve convergence in probability)

If $\mathbf{X}_n \xrightarrow{p} \mathbf{c}$ and g is continuous at $\mathbf{c}$, then $g(\mathbf{X}_n) \xrightarrow{p} g(\mathbf{c})$.

Proof. Given $\epsilon > 0$, choose $\delta > 0$ such that $\|\mathbf{x} - \mathbf{c}\| < \delta$ implies $|g(\mathbf{x}) - g(\mathbf{c})| < \epsilon$. Then $\{|g(\mathbf{X}_n) - g(\mathbf{c})| \geq \epsilon\} \subseteq \{\|\mathbf{X}_n - \mathbf{c}\| \geq \delta\}$, so

$$P(|g(\mathbf{X}_n) - g(\mathbf{c})| \geq \epsilon) \leq P(\|\mathbf{X}_n - \mathbf{c}\| \geq \delta) \to 0.$$

■

Corollary 5.8

If $X_n \xrightarrow{p} a$ and $Y_n \xrightarrow{p} b$, then $X_n \pm Y_n \xrightarrow{p} a \pm b$, $X_n Y_n \xrightarrow{p} ab$, and $X_n/Y_n \xrightarrow{p} a/b$ when $b \neq 0$.

Proof. By Theorem 5.5, $(X_n, Y_n)' \xrightarrow{p} (a, b)'$. The functions $x \pm y$, xy and x/y (for $y \neq 0$) are continuous at (a, b); apply Theorem 5.7. ■

Example 5.6 (Consistency of S_n^2 and S_n)

Let $X_1, \ldots, X_n$ be i.i.d. with $EX_1^2 < \infty$ and $\sigma^2 = \operatorname{Var} X_1 > 0$. The sample variance is a function of the first two sample moments:

$$S_n^2 = \frac{1}{n-1}\sum (X_i - \bar{X})^2 = \frac{n}{n-1}(\hat{m}_2 - \hat{m}_1^2).$$

Since $\hat{m}_1 \xrightarrow{p} m_1$, $\hat{m}_2 \xrightarrow{p} m_2$ and $n/(n-1) \to 1$, Corollary 5.8 gives $S_n^2 \xrightarrow{p} m_2 - m_1^2 = \sigma^2$. Since $\sqrt{x}$ is continuous, $S_n \xrightarrow{p} \sigma$ as well.

Theorem 5.9

If $\operatorname{Var} X_n < \infty$, $EX_n \to a$ and $\operatorname{Var} X_n \to 0$, then $X_n \xrightarrow{p} a$.

Proof. By Markov's inequality applied to $(X_n - a)^2$,

$$P(|X_n - a| \geq \epsilon) \leq \frac{E(X_n - a)^2}{\epsilon^2} = \frac{\operatorname{Var} X_n + (EX_n - a)^2}{\epsilon^2} \to 0.$$

■

Example 5.7

For uniform order statistics, $U_{(1)} \sim \text{Beta}(1, n)$, so $EU_{(1)} = 1/(n+1) \to 0$ and $\operatorname{Var} U_{(1)} = n/\{(n+1)^2(n+2)\} \to 0$. Hence $U_{(1)} \xrightarrow{p} 0$, and similarly $U_{(n)} \xrightarrow{p} 1$. More generally, for a sample from $f(x;\theta) = e^{-(x-\theta)}$, $x > \theta$, the minimum $X_{(1)} = \theta + E_{(1)}$ with $E_{(1)} \sim \text{Exp}(1/n)$, so $X_{(1)} \xrightarrow{p} \theta$.

Example 5.8 (Consistency of sample quantiles)

Let F be continuous with inverse F^{-1}, fix $0 < \alpha < 1$, and let r_n be integers with $r_n/n \to \alpha$. The order statistic $X_{(r_n)}$ is a **sample α-quantile**. We show $X_{(r_n)} \xrightarrow{p} F^{-1}(\alpha)$ when F^{-1} is continuous at α.
By Corollary 4.14 and Example 4.12, with $Z_1, \ldots, Z_n \overset{\text{i.i.d.}}{\sim} \text{Exp}(1)$,

$$X_{(r_n)} \overset{d}{=} h(Y_n), \qquad Y_n = \frac{Z_1}{n} + \frac{Z_2}{n-1} + \cdots + \frac{Z_{r_n}}{n - r_n + 1}, \qquad h(y) = F^{-1}(1 - e^{-y}).$$

Indeed $1-e^{-E}$ is uniform for $E\sim\text{Exp}(1)$, so $U_{(r)}\overset{d}{=}1-e^{-E_{(r)}}$. Now, by Riemann sums,

$$EY_n=\sum_{i=1}^{r_n}\frac{1}{n-i+1}=\frac{1}{n}\sum_{k=0}^{r_n-1}\frac{1}{1-k/n}\to\int_0^{\alpha}\frac{dx}{1-x}=-\log(1-\alpha),$$

$$\text{Var}\,Y_n=\sum_{i=1}^{r_n}\frac{1}{(n-i+1)^2}=\frac{1}{n}\cdot\frac{1}{n}\sum_{k=0}^{r_n-1}\frac{1}{(1-k/n)^2}\approx\frac{1}{n}\cdot\frac{\alpha}{1-\alpha}\to 0.$$

By Theorem 5.9, $Y_n\xrightarrow{p}-\log(1-\alpha)$. Since h is continuous at $-\log(1-\alpha)$, Theorem 5.7 gives $h(Y_n)\xrightarrow{p}h(-\log(1-\alpha))=F^{-1}(\alpha)$. Convergence in probability to a constant depends only on the distributions, so $X_{(r_n)}\xrightarrow{p}F^{-1}(\alpha)$.

5.3 Computing limiting distributions

Motivation. In practice limits are obtained by combining a few tools: the CLT, Slutsky's theorem, the continuous mapping theorem and the delta method. Together they give the asymptotic distribution of almost any statistic built from sample means.

Theorem 5.10 (Slutsky)

If $X_n\xrightarrow{d}Z$ and $Y_n\xrightarrow{p}c$ for a constant c, then

(a) $X_n+Y_n\xrightarrow{d}Z+c$;

(b) $X_n-Y_n\xrightarrow{d}Z-c$;

(c) $Y_nX_n\xrightarrow{d}cZ$;

(d) $X_n/Y_n\xrightarrow{d}Z/c$ if $c\neq 0$.

Proof. The theorem is true in general; we prove it when F_Z is continuous, which covers our applications.

(a) For $\epsilon>0$,

$P(X_n+Y_n\le z)\le P(X_n+Y_n\le z,\ |Y_n-c|<\epsilon)+P(|Y_n-c|\ge\epsilon)\le P(X_n\le z-c+\epsilon)+P(|Y_n-c|\ge\epsilon)$.

Letting $n\to\infty$, $\limsup_n P(X_n+Y_n\le z)\le P(Z\le z-c+\epsilon)$. In the same way,

$$P(X_n+Y_n>z)\le P(X_n>z-c-\epsilon)+P(|Y_n-c|\ge\epsilon),$$

so $\liminf_n P(X_n+Y_n\le z)\ge P(Z\le z-c-\epsilon)$. Let $\epsilon\downarrow 0$; by continuity of F_Z both bounds tend to $P(Z+c\le z)$.

(b) $-Y_n\xrightarrow{p}-c$; apply (a).

(c) First let $c=0$. For $\epsilon>0$ and any $k>0$,

$$P(|Y_nX_n|\ge\epsilon)\le P(|X_n|>k)+P(|Y_n|\ge\epsilon/k).$$

As $n\to\infty$ the right side tends to $P(Z>k)+P(Z<-k)$ (at continuity points), and this is arbitrarily small for large k. So $Y_nX_n\xrightarrow{p}0$. For general c, write $Y_nX_n=(Y_n-c)X_n+cX_n$: the first term $\xrightarrow{p}0$ by the case just proved, and $cX_n\xrightarrow{d}cZ$ (obvious for $c\neq 0$, trivial for $c=0$). Then apply (a).

(d) $1/Y_n\xrightarrow{p}1/c$ by Theorem 5.7; apply (c). ∎

Example 5.9 (Studentized mean)

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$. The **studentized mean** satisfies

$$\frac{\bar{X}_n - \mu}{S_n/\sqrt{n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)/\sigma}{S_n/\sigma} \xrightarrow{d} \frac{Z}{1} = Z \sim N(0,1),$$

by the CLT, Example 5.6 and Slutsky. Two consequences:

- For normal samples the left side is exactly $t(n-1)$, so $t(n-1) \xrightarrow{d} N(0,1)$: for large n, t quantiles are close to normal ones.
- For any population with finite variance, $P(\bar{X}_n - z_{\alpha/2} S_n/\sqrt{n} \le \mu \le \bar{X}_n + z_{\alpha/2} S_n/\sqrt{n}) \to 1-\alpha$, so $\bar{X}_n \pm z_{\alpha/2} S_n/\sqrt{n}$ is an **asymptotic** $100(1-\alpha)\%$ confidence interval for μ.

For instance, if $n = 100$ waiting times give $\bar{x} = 12.4$ and $s = 3.0$, the asymptotic 95% interval is $12.4 \pm 1.96 \cdot 0.3 = (11.81, 12.99)$, whatever the shape of the population.

Example 5.10 (Sample variance)

Let $\mu_4 = E(X_1 - \mu)^4 < \infty$. Write

$$S_n^2 = \frac{n}{n-1}\Big\{\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 - (\bar{X}_n - \mu)^2\Big\}.$$

The variables $Y_i = (X_i - \mu)^2$ are i.i.d. with mean σ^2 and variance $\mu_4 - \sigma^4$, so by the CLT $\sqrt{n}(\frac{1}{n}\sum Y_i - \sigma^2) \xrightarrow{d} N(0, \mu_4 - \sigma^4)$. Also $\sqrt{n}(\bar{X}_n - \mu)^2 = \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu) \xrightarrow{d} 0$ by Slutsky (a $\xrightarrow{d} N(0, \sigma^2)$ factor times a $\xrightarrow{p} 0$ factor). Therefore

$$\sqrt{n}(S_n^2 - \sigma^2) = \frac{n}{n-1}\Big[\sqrt{n}\Big(\frac{1}{n}\sum Y_i - \sigma^2\Big) - \sqrt{n}(\bar{X}_n - \mu)^2\Big] + \frac{\sqrt{n}}{n-1}\sigma^2 \xrightarrow{d} N(0, \mu_4 - \sigma^4).$$

With the excess kurtosis $\rho_4 = \mu_4/\sigma^4 - 3$, the limit variance is $(\rho_4 + 2)\sigma^4$. For normal populations $\rho_4 = 0$ and the variance is $2\sigma^4$, in line with $\operatorname{Var} S_n^2 = 2\sigma^4/(n-1)$.

Definition (Convergence in distribution of vectors)

$\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$ if $P(\mathbf{X}_n \le \mathbf{x}) \to P(\mathbf{Z} \le \mathbf{x})$ at every $\mathbf{x}$ where the joint cdf of $\mathbf{Z}$ is continuous.

Theorem 5.11 (Continuous mapping)

If $\mathbf{X}_n \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$ and g is continuous, then $g(\mathbf{X}_n) \xrightarrow{d} g(\mathbf{Z})$.

The proof is beyond the level of these notes. Note that Slutsky's theorem does *not* follow from this unless the pair (X_n, Y_n) converges jointly. Joint convergence holds when one component converges to a constant, and that is exactly Slutsky's setting.

Example 5.11 (Binomial and chi-square)

If $X_n \sim B(n,p)$, then $(X_n - np)/\sqrt{np(1-p)} \xrightarrow{d} Z$, and squaring (continuous) gives

$\frac{(X_n-np)^2}{np(1-p)} \xrightarrow{d} \chi^2(1)$. With two cells $X_{n1} = X_n$, $X_{n2} = n - X_n$ and $p_1 = p$, $p_2 = 1-p$,

$$\frac{(X_{n1}-np_1)^2}{np_1} + \frac{(X_{n2}-np_2)^2}{np_2} = (X_n - np)^2\Big(\frac{1}{np} + \frac{1}{n(1-p)}\Big) = \frac{(X_n-np)^2}{np(1-p)}.$$

So the two-cell Pearson statistic is asymptotically $\chi^2(1)$.

Example 5.12 (Pearson's chi-square statistic)

Let $(X_{n1},\dots,X_{nk})' \sim \text{Multi}(n,(p_1,\dots,p_k)')$, all $p_j > 0$. We show

$$Q_n = \sum_{j=1}^{k} \frac{(X_{nj}-np_j)^2}{np_j} \xrightarrow{d} \chi^2(k-1).$$

Drop the last cell: let $\mathbf{Y}_n = (X_{n1},\dots,X_{n,k-1})'$, $\mu = (p_1,\dots,p_{k-1})'$ and $r = k-1$. By Example 5.2, $(\mathbf{Y}_n - n\mu)/\sqrt{n} \xrightarrow{d} \mathbf{Z} \sim N_r(\mathbf{0},\Sigma)$ with $\Sigma = D(\mu) - \mu\mu'$, which is now nonsingular. The quadratic form $\mathbf{z}'\Sigma^{-1}\mathbf{z}$ is continuous, so by Theorem 5.11 and Theorem 4.18(a),

$$\frac{1}{n}(\mathbf{Y}_n - n\mu)'\Sigma^{-1}(\mathbf{Y}_n - n\mu) \xrightarrow{d} \mathbf{Z}'\Sigma^{-1}\mathbf{Z} \sim \chi^2(k-1).$$

It remains to identify the left side with Q_n. One checks directly that

$$\Sigma^{-1} = D(\mu)^{-1} + \frac{1}{p_k}\mathbf{1}\mathbf{1}' :$$

multiplying, $(D-\mu\mu')(D^{-1}+\mathbf{1}\mathbf{1}'/p_k) = I + \mu\mathbf{1}'/p_k - \mu\mathbf{1}' - (1-p_k)\mu\mathbf{1}'/p_k = I$, using $D\mathbf{1} = \mu$ and $\mu'\mathbf{1} = 1-p_k$. Hence, with $d_j = X_{nj} - np_j$ and $\sum_{j<k} d_j = -d_k$,

$$\frac{1}{n}(\mathbf{Y}_n - n\mu)'\Sigma^{-1}(\mathbf{Y}_n - n\mu) = \sum_{j=1}^{k-1}\frac{d_j^2}{np_j} + \frac{(\sum_{j<k} d_j)^2}{np_k} = \sum_{j=1}^{k}\frac{d_j^2}{np_j} = Q_n.$$

Theorem 5.12 (Delta method)

Suppose $\sqrt{n}(\mathbf{X}_n - \theta) \xrightarrow{d} \mathbf{Z}$ in $\mathbb{R}^k$, and $g:\mathbb{R}^k \to \mathbb{R}$ has continuous first partial derivatives near θ, with gradient $\dot g(\theta) = (\partial g/\partial\theta_1,\dots,\partial g/\partial\theta_k)'$. Then

$$\sqrt{n}(g(\mathbf{X}_n) - g(\theta)) \xrightarrow{d} \dot g(\theta)'\mathbf{Z}.$$

In particular, if $\mathbf{Z} \sim N_k(\mathbf{0},\Sigma)$, the limit is $N\big(0, \dot g(\theta)'\Sigma\dot g(\theta)\big)$.

Proof. We give the one-dimensional proof; the multivariate one is the same. Since g is differentiable at θ, there is a function r with $r(x) \to 0$ as $x \to \theta$ such that

$$g(x) = g(\theta) + \{\dot g(\theta) + r(x)\}(x-\theta).$$

Set $r(\theta) = 0$. First, $X_n \xrightarrow{p} \theta$: by Slutsky, $X_n - \theta = \frac{1}{\sqrt{n}}\cdot\sqrt{n}(X_n-\theta) \xrightarrow{d} 0\cdot Z = 0$, and Theorem 5.3 applies. Next, $r(X_n) \xrightarrow{p} 0$: given $\epsilon > 0$ choose δ with $|x-\theta| < \delta \Rightarrow |r(x)| < \epsilon$; then $P(|r(X_n)| \ge \epsilon) \le P(|X_n - \theta| \ge \delta) \to 0$. Finally

$$\sqrt{n}(g(X_n) - g(\theta)) = \{\dot g(\theta) + r(X_n)\}\sqrt{n}(X_n - \theta),$$

where the first factor $\xrightarrow{p} \dot g(\theta)$ and the second $\xrightarrow{d} Z$. Slutsky's theorem gives the limit $\dot g(\theta)Z$. ∎

Example 5.13 (Sample standard deviation)

From Example 5.10, $\sqrt{n}(S_n^2 - \sigma^2) \xrightarrow{d} N(0, (\rho_4 + 2)\sigma^4)$. With $g(x) = \sqrt{x}$, $\dot{g}(\sigma^2) = 1/(2\sigma)$, so

$$\sqrt{n}(S_n - \sigma) \xrightarrow{d} N\Big(0, \frac{(\rho_4 + 2)\sigma^2}{4}\Big).$$

For normal data this is $N(0, \sigma^2/2)$.

Example 5.14 (Sample correlation coefficient)

Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be i.i.d. pairs with correlation ρ and finite fourth moments, and

$$\hat{\rho}_n = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2}\sqrt{\sum (Y_i - \bar{Y})^2}}.$$

$\hat{\rho}_n$ is unchanged if we standardize each coordinate, so we may assume $\mu_1 = \mu_2 = 0$, $\sigma_1 = \sigma_2 = 1$. Write the five sample means $\bar{X}, \bar{Y}, \overline{XY}, \overline{X^2}, \overline{Y^2}$ as the average of $\mathbf{W}_i = (X_i, Y_i, X_iY_i, X_i^2, Y_i^2)'$. Then $\hat{\rho}_n = g(\bar{\mathbf{W}}_n)$ with

$$g(t_1, \ldots, t_5) = \frac{t_3 - t_1 t_2}{\sqrt{t_4 - t_1^2}\sqrt{t_5 - t_2^2}}, \qquad \theta = E\mathbf{W}_1 = (0, 0, \rho, 1, 1)', \qquad g(\theta) = \rho.$$

The gradient at θ is $\dot{g}(\theta) = (0, 0, 1, -\rho/2, -\rho/2)'$. By the multivariate CLT and Theorem 5.12,

$$\sqrt{n}(\hat{\rho}_n - \rho) \xrightarrow{d} N\Big(0,\ \mathrm{Var}\Big(X_1Y_1 - \frac{\rho}{2}X_1^2 - \frac{\rho}{2}Y_1^2\Big)\Big).$$

Bivariate normal case. Write $Y_1 = \rho X_1 + \sqrt{1 - \rho^2}\, T$ with X_1, T independent $N(0,1)$ (Theorem 4.17). Expanding,

$$X_1Y_1 - \frac{\rho}{2}X_1^2 - \frac{\rho}{2}Y_1^2 = (1 - \rho^2)\Big\{\frac{\rho}{2}(X_1^2 - T^2) + \sqrt{1 - \rho^2}\, X_1 T\Big\}.$$

Here $\mathrm{Var}(X_1^2 - T^2) = 4$, $\mathrm{Var}(X_1T) = 1$, and the two terms are uncorrelated. So the variance is $(1 - \rho^2)^2\{\rho^2 + 1 - \rho^2\} = (1 - \rho^2)^2$:

$$\sqrt{n}(\hat{\rho}_n - \rho) \xrightarrow{d} N(0, (1 - \rho^2)^2).$$

Example 5.15 (Variance-stabilizing transformation)

In Example 5.14 the limit variance depends on the unknown ρ. For a function g, the delta method gives $\sqrt{n}(g(\hat{\rho}_n) - g(\rho)) \xrightarrow{d} N(0, \dot{g}(\rho)^2(1 - \rho^2)^2)$. Choosing $\dot{g}(\rho) = 1/(1 - \rho^2)$, i.e. **Fisher's z-transform**

$$g(\rho) = \frac{1}{2}\log\frac{1 + \rho}{1 - \rho} = \tanh^{-1}\rho,$$

makes the limit $N(0,1)$, free of ρ. Such a g is a **variance-stabilizing transformation**. The asymptotic interval $\tanh^{-1}\hat{\rho}_n \pm z_{\alpha/2}/\sqrt{n}$ for $\tanh^{-1}\rho$ maps back to an interval for ρ. With $n = 100$ and $\hat{\rho} = 0.6$: $\tanh^{-1} 0.6 = 0.693$, the interval is $0.693 \pm 0.196 = (0.497, 0.889)$, and applying tanh gives $(0.460, 0.711)$ for ρ.

Example 5.16 (Asymptotic normality of sample quantiles)

In the setting of Example 5.8, assume F has a density f with $f(\xi_\alpha) > 0$ at $\xi_\alpha = F^{-1}(\alpha)$, and $r_n/n = \alpha + o(n^{-1/2})$. Then

$$\sqrt{n}(X_{(r_n)} - \xi_\alpha) \xrightarrow{d} N\Big(0, \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2}\Big).$$

Sketch. With Y_n as in Example 5.8, its cgf is $K(s) = -\sum_{i=1}^{r_n} \log\left(1 - \frac{s}{n-i+1}\right) = \sum_{m\geq 1} \frac{s^m}{m} \sum_i (n-i+1)^{-m}$. Taking $s = \sqrt{n}\, t/\tau$ with $\tau^2 = \alpha/(1-\alpha)$, the $m = 1$ term gives the centring, the $m = 2$ term tends to $t^2/2$, and the terms with $m \geq 3$ are $O(n^{m/2} \cdot n^{1-m}) \to 0$. Hence $\sqrt{n}(Y_n + \log(1-\alpha)) \xrightarrow{d} N(0, \alpha/(1-\alpha))$. Next apply the delta method to $h(y) = F^{-1}(1 - e^{-y})$, whose derivative at $y_0 = -\log(1-\alpha)$ is

$$\dot{h}(y_0) = \frac{e^{-y_0}}{f(F^{-1}(1-e^{-y_0}))} = \frac{1-\alpha}{f(\xi_\alpha)}.$$

The limit variance is $\frac{(1-\alpha)^2}{f(\xi_\alpha)^2} \cdot \frac{\alpha}{1-\alpha} = \frac{\alpha(1-\alpha)}{f(\xi_\alpha)^2}$.
For the sample median ($\alpha = 1/2$), the variance is $1/\{4f(m)^2\}$. For $N(\mu, \sigma^2)$ data, $f(\mu) = 1/(\sigma\sqrt{2\pi})$, so the median has asymptotic variance $\pi\sigma^2/2$, against σ^2 for the mean: the mean is more efficient by the factor $\pi/2 \approx 1.57$. For double exponential data $\frac{1}{2}e^{-|x-\mu|}$, the median has variance 1 against 2 for the mean, and the order is reversed.

5.4 Approximation by simulation

Values generated by a computer to imitate observations of a random variable are called **random numbers**; those imitating $U(0,1)$ are **uniform random numbers**. By the probability integral transform (Theorem 4.13), $F^{-1}(U) \sim F$, so uniform random numbers give random numbers from any distribution.

Example 5.17 (Generating random numbers)

- Logistic: $F(z) = 1/(1+e^{-z})$ has $F^{-1}(u) = \log\frac{u}{1-u}$, so $\log\frac{U}{1-U} \sim L(0,1)$, and $\mu + \sigma\log\frac{U}{1-U} \sim L(\mu,\sigma)$.
- Exponential: $-\beta\log(1-U) \sim \mathrm{Exp}(\beta)$ (and so is $-\beta\log U$).
- Cauchy: $\tan\{\pi(U - \frac{1}{2})\}$ is standard Cauchy.
- Normal: $\mu + \sigma\Phi^{-1}(U) \sim N(\mu, \sigma^2)$; alternatively use the polar method of Example 4.6.

Sums and ratios of such variates give gamma, χ^2, t and F random numbers through the representations of Chapter 4.

Theorem 5.13 (Monte Carlo integration)

Let g be continuous on $[a,b]$ and $U_1, \ldots, U_n \overset{\text{i.i.d.}}{\sim} U(0,1)$. With $X_i = a + (b-a)U_i \sim U(a,b)$,

$$\frac{b-a}{n}\sum_{i=1}^{n} g(X_i) \xrightarrow{p} \int_a^b g(x)\,dx.$$

Proof. X_i has pdf $\frac{1}{b-a}$ on (a,b), so $Eg(X_1) = \frac{1}{b-a}\int_a^b g(x)\,dx$, finite since g is bounded. By the law of large numbers, $\frac{b-a}{n}\sum g(X_i) \xrightarrow{p} (b-a)Eg(X_1) = \int_a^b g$. ∎

So with uniform random numbers $u_1, \dots, u_n$ and $y_i = (b-a)g(a+(b-a)u_i)$, the average $\bar{y}$ approximates the integral. Since $\bar{y}$ is a sample mean, Example 5.9 gives the asymptotic 95% interval $\bar{y} \pm 1.96 s_y/\sqrt{n}$ as an error bound; the error shrinks like $1/\sqrt{n}$.

Example 5.18

To approximate $\int_0^2 x^3\,dx = 4$, use $y_i = 2(2u_i)^3 = 16u_i^3$. Here $\mathrm{Var}(16U^3) = 256(\frac{1}{7} - \frac{1}{16}) = 20.6$, so with $n = 10{,}000$ the standard error is about $\sqrt{20.6}/100 = 0.045$, and the 95% error bound is about ± 0.09. Quadrupling n halves it.

Example 5.19 (Variance of the logistic distribution)

The standard logistic distribution is symmetric about 0 with variance $\int x^2 e^{-x}(1+e^{-x})^{-2}dx = \pi^2/3 \approx 3.290$. (Expand $(1+e^{-x})^{-2} = \sum_{m\geq 1}(-1)^{m-1} m e^{-(m-1)x}$ on $x > 0$ to get $4\sum_{m\geq 1}(-1)^{m-1}/m^2 = 4 \cdot \pi^2/12$.) To check this by simulation, take $x_i = \log \frac{u_i}{1-u_i}$ and average $y_i = x_i^2$; the estimate with its interval settles near 3.29 as n grows.

Example 5.20 (Estimating π)

If $U_1, U_2 \overset{\text{i.i.d.}}{\sim} U(0,1)$, the point (U_1, U_2) is uniform on the unit square and $p = P(U_1^2 + U_2^2 \leq 1) = \pi/4$. With n independent points, let $\hat{p}_n$ be the proportion inside the quarter disk. By the LLN, $4\hat{p}_n \xrightarrow{p} \pi$, and by the CLT and Slutsky,

$$\frac{\hat{p}_n - p}{\sqrt{\hat{p}_n(1-\hat{p}_n)/n}} \xrightarrow{d} N(0,1),$$

so $4\hat{p}_n \pm 4z_{\alpha/2}\sqrt{\hat{p}_n(1-\hat{p}_n)/n}$ is an asymptotic interval for π. Since $4\sqrt{p(1-p)} \approx 1.64$, with $n = 10{,}000$ the 95% half-width is about $1.96 \cdot 1.64/100 \approx 0.032$: two correct decimals need about 10^4 points, and each further digit costs a factor of 100.